The Fractal Five · Year 10 · gallery of new items

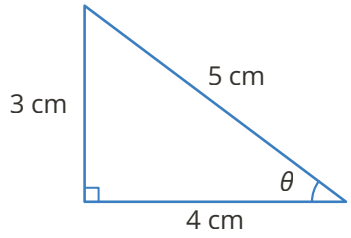
Every new pool item as it prints (figures at the printed size), with the answer computed from the same numbers that label the figure and the formula printed in Terms 1–2 (and for the Term 4 pyramid and cone). L1 = Terms 1–2 pool; L2 = Terms 3–4 pool (Terms 3–4 also draw on half of L1).

A RIGHT-ANGLED TRIANGLES AND SIMILAR FIGURES

trig_ratio L1 · (3, 4, 5, 'cm', 'sin', 0)

Write sin θ as a fraction.

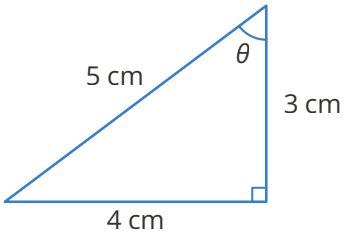
→ **3/5** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (4, 3, 5, 'cm', 'cos', 1)

Write cos θ as a fraction.

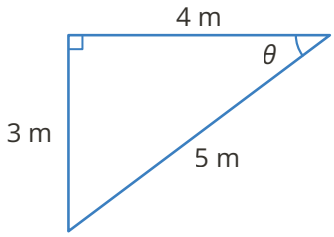
→ **3/5** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (3, 4, 5, 'm', 'tan', 2)

Write tan θ as a fraction.

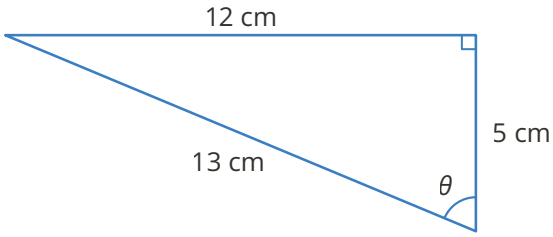
→ **3/4** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (12, 5, 13, 'cm', 'sin', 3)

Write sin θ as a fraction.

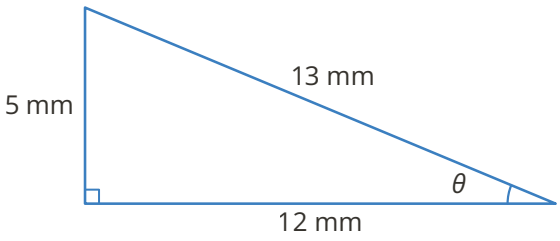
→ **12/13** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (5, 12, 13, 'mm', 'cos', 0)

Write cos θ as a fraction.

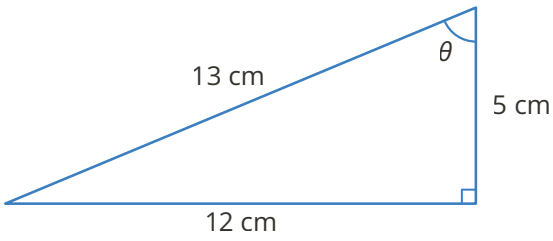
→ **12/13** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (12, 5, 13, 'cm', 'tan', 1)

Write tan θ as a fraction.

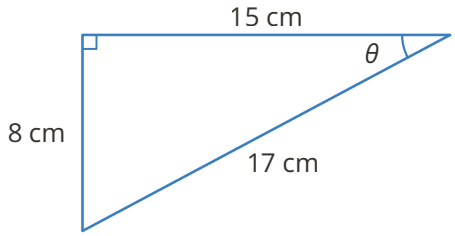
→ **12/5** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (8, 15, 17, 'cm', 'sin', 2)

Write sin θ as a fraction.

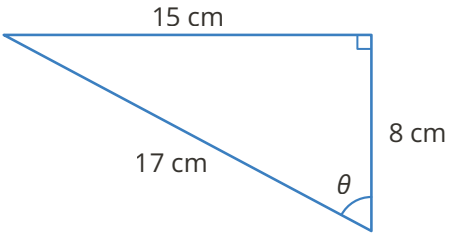
→ **8/17** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (15, 8, 17, 'cm', 'cos', 3)

Write cos θ as a fraction.

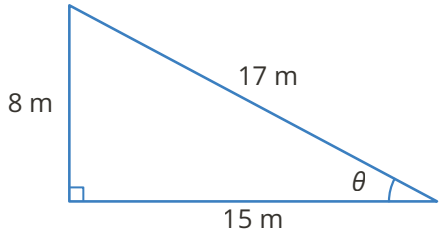
→ **8/17** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (8, 15, 17, 'm', 'tan', 0)

Write tan θ as a fraction.

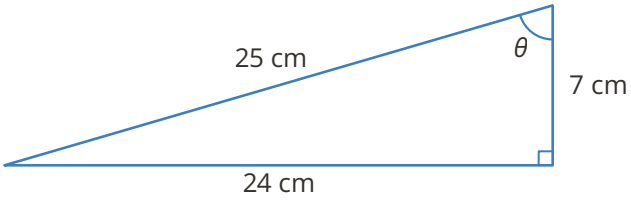
→ **8/15** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (24, 7, 25, 'cm', 'sin', 1)

Write sin θ as a fraction.

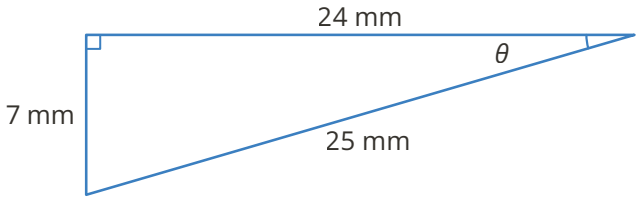
→ **24/25** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (7, 24, 25, 'mm', 'cos', 2)

Write cos θ as a fraction.

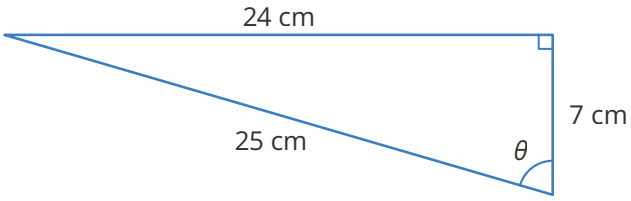
→ **24/25** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L1 · (24, 7, 25, 'cm', 'tan', 3)

Write tan θ as a fraction.

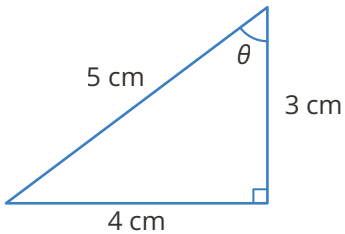
→ **24/7** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (4, 3, 5, 'cm', 'sin', 1)

Write sin θ as a fraction.

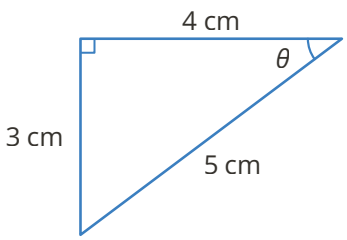
→ **4/5** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (3, 4, 5, 'cm', 'cos', 2)

Write cos θ as a fraction.

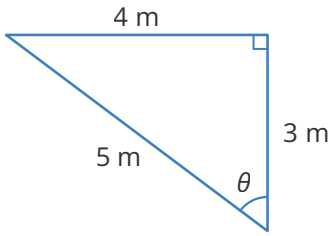
→ **4/5** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (4, 3, 5, 'm', 'tan', 3)

Write tan θ as a fraction.

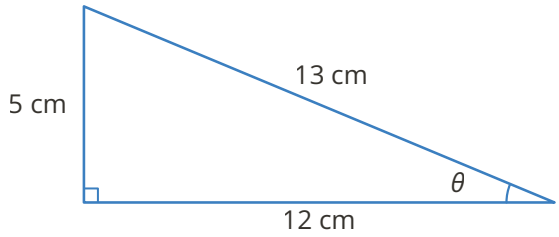
→ **4/3** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (5, 12, 13, 'cm', 'sin', 0)

Write sin θ as a fraction.

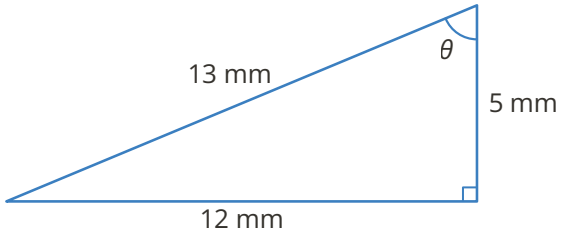
→ **5/13** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (12, 5, 13, 'mm', 'cos', 1)

Write cos θ as a fraction.

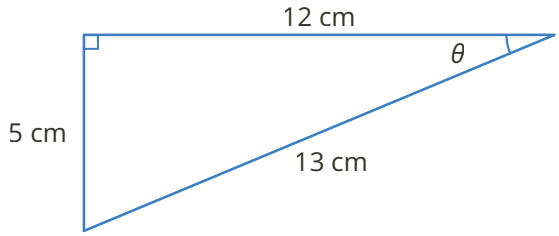
→ **5/13** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (5, 12, 13, 'cm', 'tan', 2)

Write tan θ as a fraction.

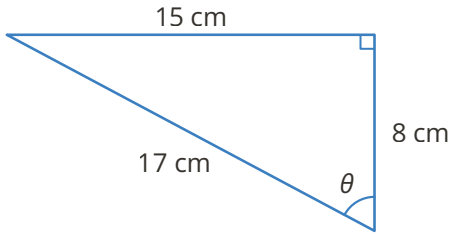
→ **5/12** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (15, 8, 17, 'cm', 'sin', 3)

Write sin θ as a fraction.

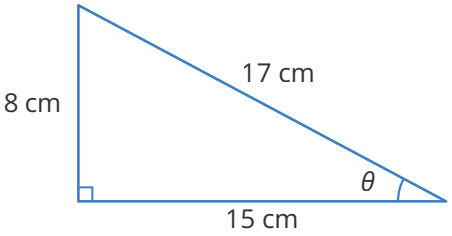
→ **15/17** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (8, 15, 17, 'cm', 'cos', 0)

Write cos θ as a fraction.

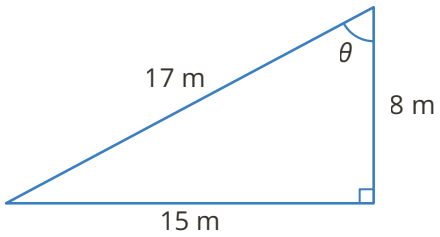
→ **15/17** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (15, 8, 17, 'm', 'tan', 1)

Write tan θ as a fraction.

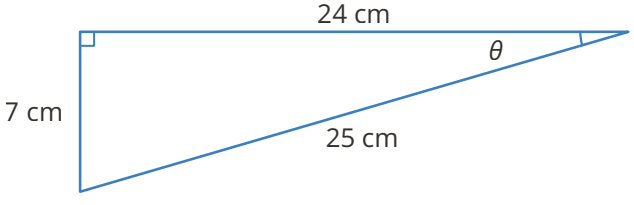
→ **15/8** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (7, 24, 25, 'cm', 'sin', 2)

Write sin θ as a fraction.

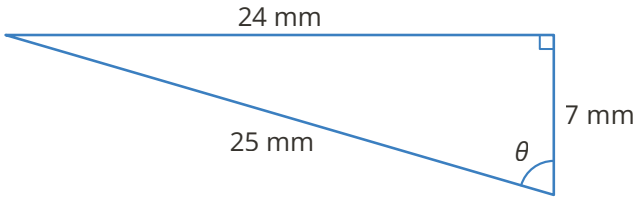
→ **7/25** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (24, 7, 25, 'mm', 'cos', 3)

Write cos θ as a fraction.

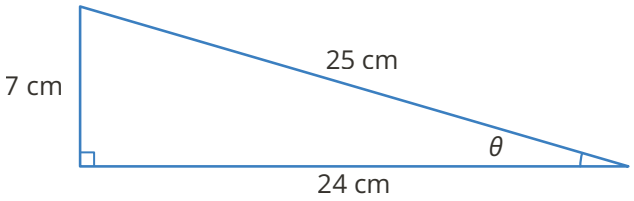
→ **7/25** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_ratio L2 · (7, 24, 25, 'cm', 'tan', 0)

Write tan θ as a fraction.

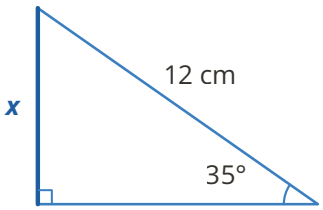
→ **7/24** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('hyp', 12, 35, 'opp', 'cm', 0)

Find *x*, correct to 1 decimal place.

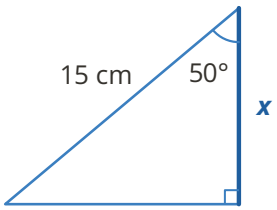
→ **6.9 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('hyp', 15, 50, 'adj', 'cm', 1)

Find *x*, correct to 1 decimal place.

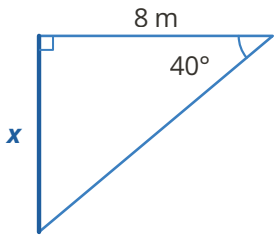
→ **9.6 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('adj', 8, 40, 'opp', 'm', 2)

Find *x*, correct to 1 decimal place.

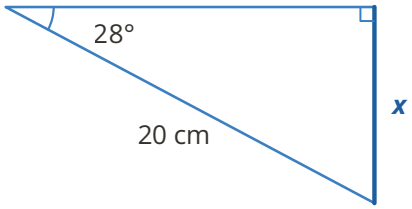
→ **6.7 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('hyp', 20, 28, 'opp', 'cm', 3)

Find *x*, correct to 1 decimal place.

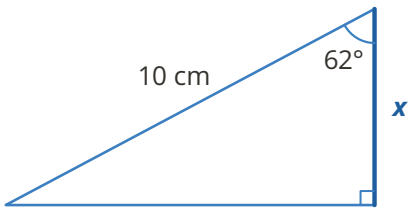
→ **9.4 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('hyp', 10, 62, 'adj', 'cm', 1)

Find *x*, correct to 1 decimal place.

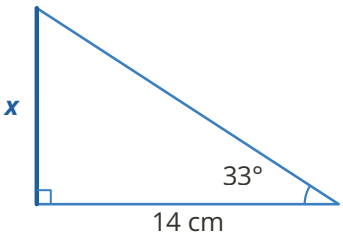
→ **4.7 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('adj', 14, 33, 'opp', 'cm', 0)

Find *x*, correct to 1 decimal place.

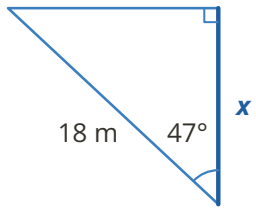
→ **9.1 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('hyp', 18, 47, 'adj', 'm', 3)

Find *x*, correct to 1 decimal place.

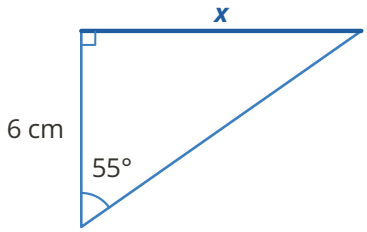
→ **12.3 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('adj', 6, 55, 'opp', 'cm', 2)

Find *x*, correct to 1 decimal place.

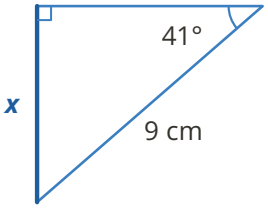
→ **8.6 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('hyp', 9, 41, 'opp', 'cm', 2)

Find *x*, correct to 1 decimal place.

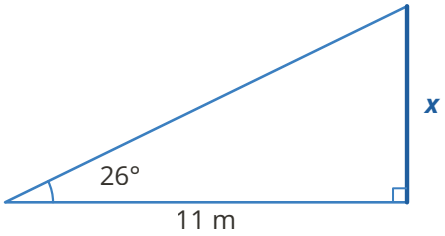
→ **5.9 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L1 · ('adj', 11, 26, 'opp', 'm', 1)

Find *x*, correct to 1 decimal place.

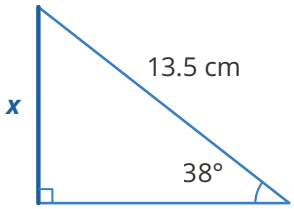
→ **5.4 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · ('hyp', 13.5, 38, 'opp', 'cm', 0)

Find *x*, correct to 1 decimal place.

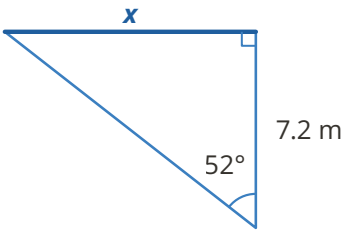
→ **8.3 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · ('adj', 7.2, 52, 'opp', 'm', 3)

Find *x*, correct to 1 decimal place.

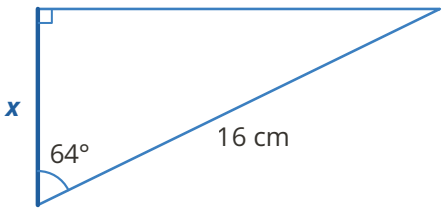
→ **9.2 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · ('hyp', 16, 64, 'adj', 'cm', 2)

Find *x*, correct to 1 decimal place.

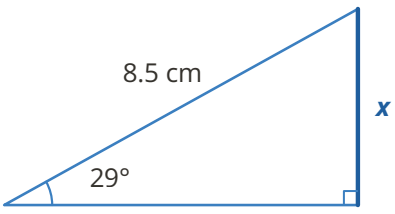
→ **7.0 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · ('hyp', 8.5, 29, 'opp', 'cm', 1)

Find *x*, correct to 1 decimal place.

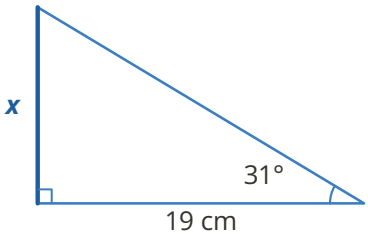
→ **4.1 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · ('adj', 19, 31, 'opp', 'cm', 0)

Find *x*, correct to 1 decimal place.

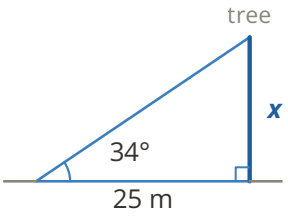
→ **11.4 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · elevation/depression (T3-4) · ('elev', 'tree', 25, 34, 'm', 1)

The angle of elevation is 34°. Find *x*, correct to 1 decimal place.

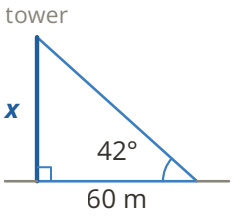
→ **16.9 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · elevation/depression (T3-4) · ('elev', 'tower', 60, 42, 'm', 0)

The angle of elevation is 42°. Find *x*, correct to 1 decimal place.

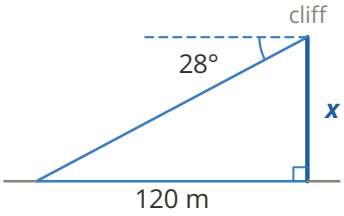
→ **54.0 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · elevation/depression (T3-4) · ('dep', 'cliff', 120, 28, 'm', 1)

The angle of depression is 28°. Find *x*, correct to 1 decimal place.

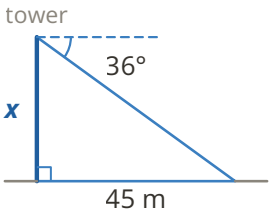
→ **63.8 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_top L2 · elevation/depression (T3-4) · ('dep', 'tower', 45, 36, 'm', 0)

The angle of depression is 36°. Find *x*, correct to 1 decimal place.

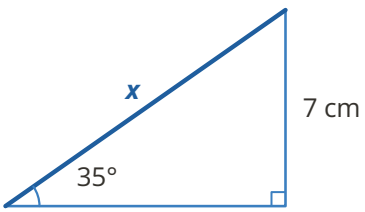
→ **32.7 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('opp', 7, 35, 'hyp', 'cm', 1)

Find *x*, correct to 1 decimal place.

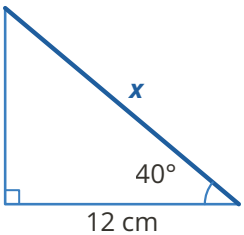
→ **12.2 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('adj', 12, 40, 'hyp', 'cm', 0)

Find *x*, correct to 1 decimal place.

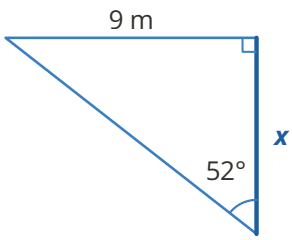
→ **15.7 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('opp', 9, 52, 'adj', 'm', 3)

Find *x*, correct to 1 decimal place.

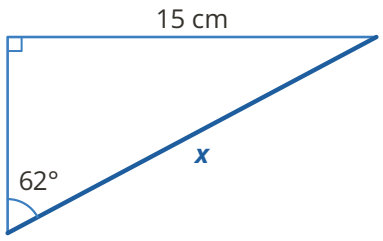
→ **7.0 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('opp', 15, 62, 'hyp', 'cm', 2)

Find *x*, correct to 1 decimal place.

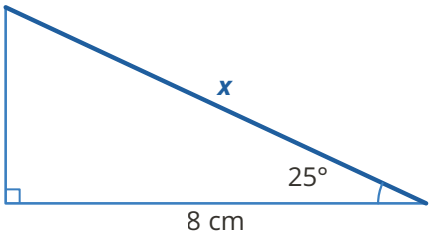
→ **17.0 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('adj', 8, 25, 'hyp', 'cm', 0)

Find *x*, correct to 1 decimal place.

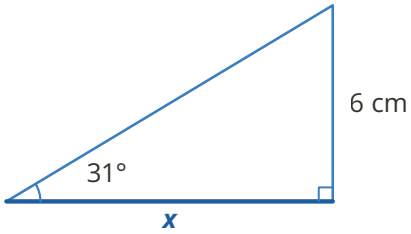
→ **8.8 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('opp', 6, 31, 'adj', 'cm', 1)

Find *x*, correct to 1 decimal place.

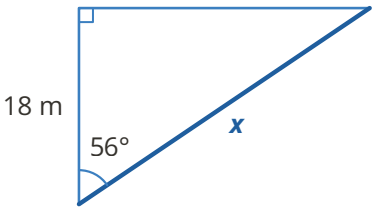
→ **10.0 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('adj', 18, 56, 'hyp', 'm', 2)

Find *x*, correct to 1 decimal place.

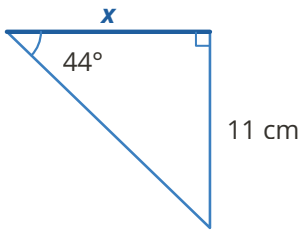
→ **32.2 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('opp', 11, 44, 'adj', 'cm', 3)

Find *x*, correct to 1 decimal place.

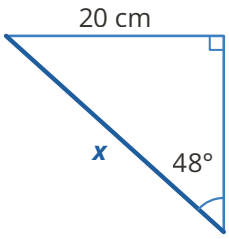
→ **11.4 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('opp', 20, 48, 'hyp', 'cm', 3)

Find *x*, correct to 1 decimal place.

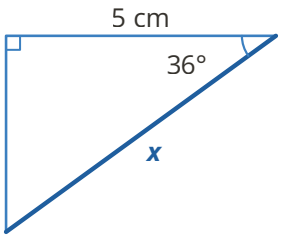
→ **26.9 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L1 · ('adj', 5, 36, 'hyp', 'cm', 2)

Find *x*, correct to 1 decimal place.

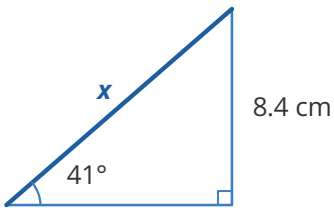
→ **6.2 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L2 · ('opp', 8.4, 41, 'hyp', 'cm', 1)

Find *x*, correct to 1 decimal place.

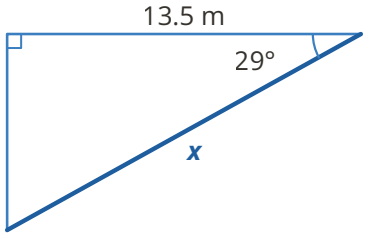
→ **12.8 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L2 · ('adj', 13.5, 29, 'hyp', 'm', 2)

Find x, correct to 1 decimal place.

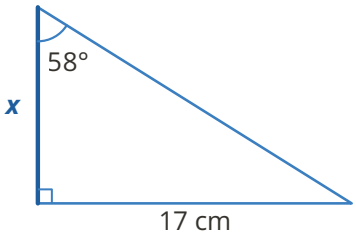
→ **15.4 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L2 · ('opp', 17, 58, 'adj', 'cm', 0)

Find x, correct to 1 decimal place.

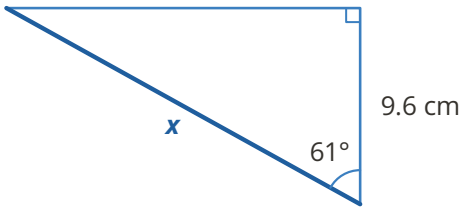
→ **10.6 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L2 · ('adj', 9.6, 61, 'hyp', 'cm', 3)

Find x, correct to 1 decimal place.

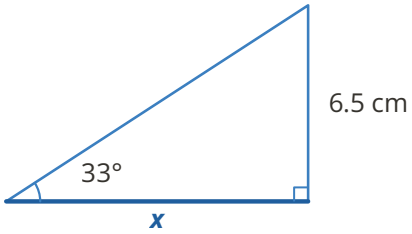
→ **19.8 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L2 · ('opp', 6.5, 33, 'adj', 'cm', 1)

Find x, correct to 1 decimal place.

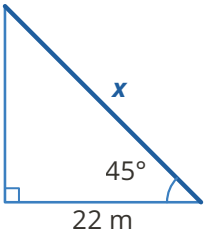
→ **10.0 cm** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_bottom L2 · ('adj', 22, 45, 'hyp', 'm', 0)

Find x, correct to 1 decimal place.

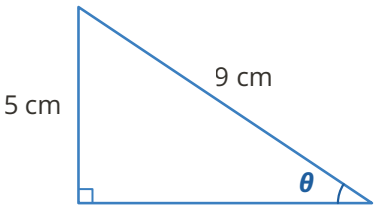
→ **31.1 m** $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$



trig_angle L1 · ('sin', 5, 9, 'cm', 0)

Find θ, correct to the nearest degree.

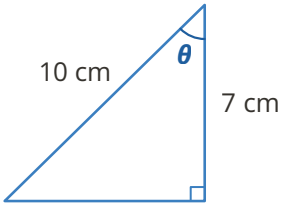
→ **34°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('cos', 7, 10, 'cm', 1)

Find θ, correct to the nearest degree.

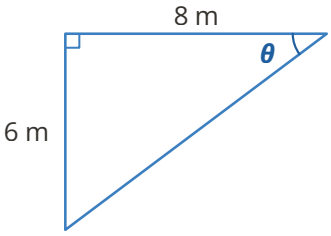
→ **46°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('tan', 6, 8, 'm', 2)

Find θ, correct to the nearest degree.

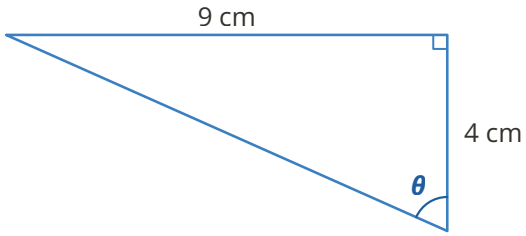
→ **37°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('tan', 9, 4, 'cm', 3)

Find θ, correct to the nearest degree.

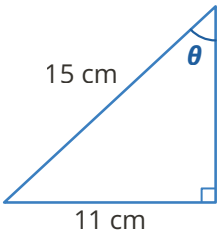
→ **66°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('sin', 11, 15, 'cm', 1)

Find θ, correct to the nearest degree.

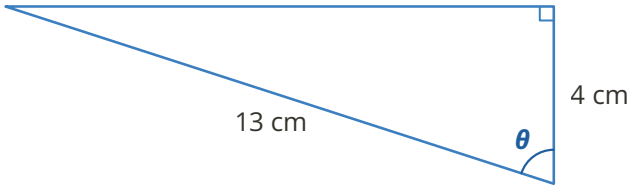
→ **47°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('cos', 4, 13, 'cm', 3)

Find θ, correct to the nearest degree.

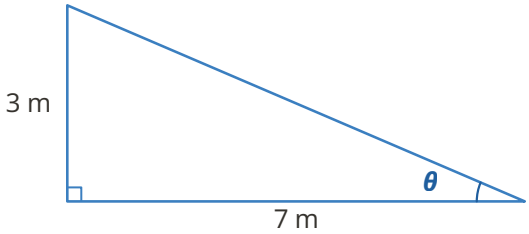
→ **72°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('tan', 3, 7, 'm', 0)

Find θ, correct to the nearest degree.

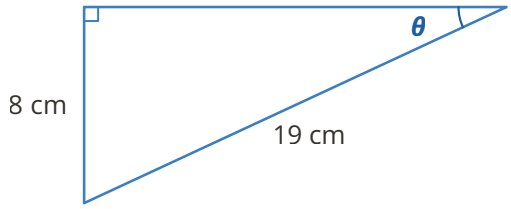
→ **23°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('sin', 8, 19, 'cm', 2)

Find θ, correct to the nearest degree.

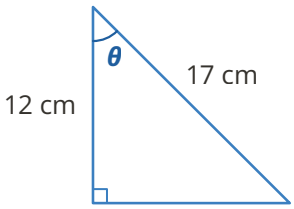
→ **25°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('cos', 12, 17, 'cm', 0)

Find θ, correct to the nearest degree.

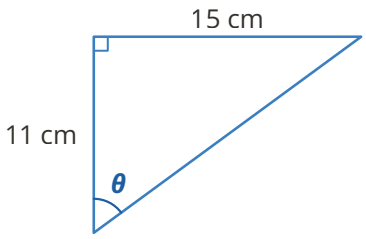
→ **45°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L1 · ('tan', 15, 11, 'cm', 2)

Find θ, correct to the nearest degree.

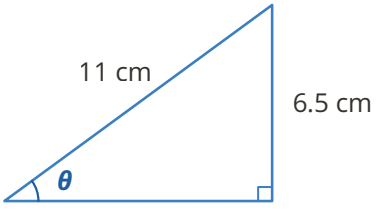
→ **54°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · ('sin', 6.5, 11, 'cm', 1)

Find θ, correct to the nearest degree.

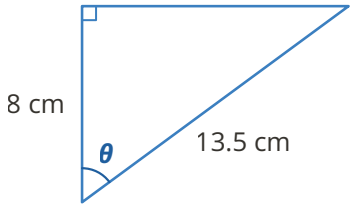
→ **36°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · ('cos', 8, 13.5, 'cm', 2)

Find θ, correct to the nearest degree.

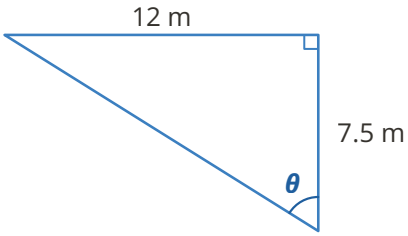
→ **54°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · ('tan', 12, 7.5, 'm', 3)

Find θ, correct to the nearest degree.

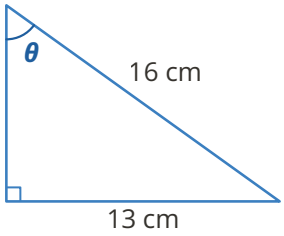
→ **58°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · ('sin', 13, 16, 'cm', 0)

Find θ, correct to the nearest degree.

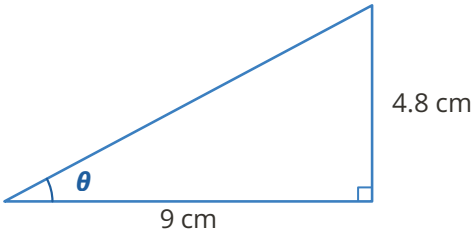
→ **54°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · ('tan', 4.8, 9, 'cm', 1)

Find θ, correct to the nearest degree.

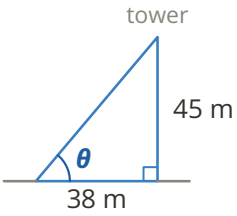
→ **28°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · elevation/depression (T3-4) · ('elev', 'tower', 45, 38, 'm', 1)

Find the angle of elevation, θ, correct to the nearest degree.

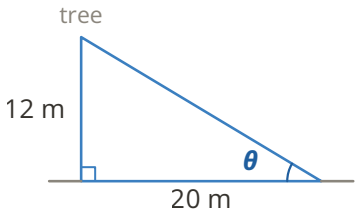
→ **50°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · elevation/depression (T3-4) · ('elev', 'tree', 12, 20, 'm', 0)

Find the angle of elevation, θ, correct to the nearest degree.

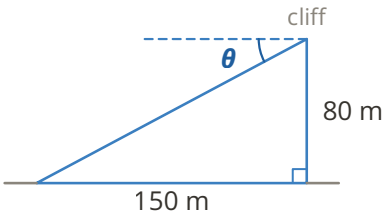
→ **31°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · elevation/depression (T3-4) · ('dep', 'cliff', 80, 150, 'm', 1)

Find the angle of depression, θ, correct to the nearest degree.

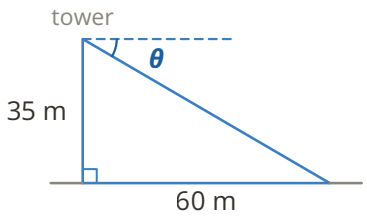
→ **28°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



trig_angle L2 · elevation/depression (T3-4) · ('dep', 'tower', 35, 60, 'm', 0)

Find the angle of depression, θ, correct to the nearest degree.

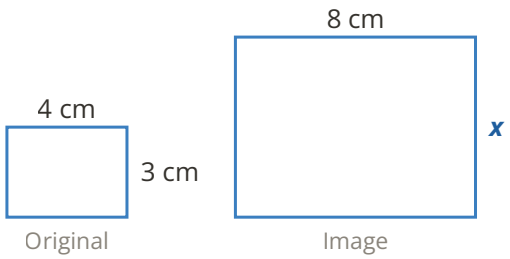
→ **30°** $\theta = \sin^{-1}(O/H)$, $\cos^{-1}(A/H)$ or $\tan^{-1}(O/A)$



similar L1 · (4, 3, 2, 'iw')

These rectangles are similar. Find *x*.

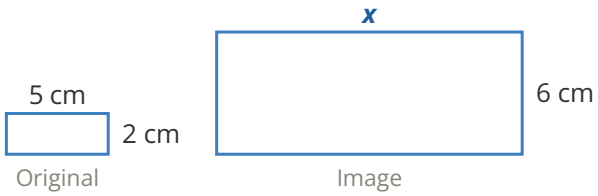
→ **6 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (5, 2, 3, 'il')

These rectangles are similar. Find *x*.

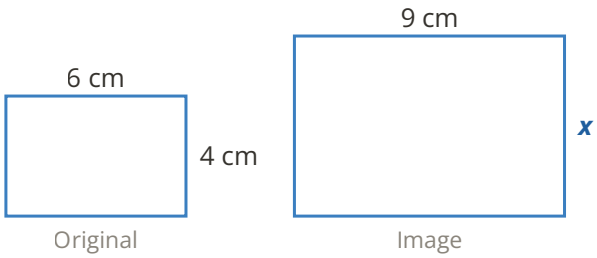
→ **15 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (6, 4, 1.5, 'iw')

These rectangles are similar. Find *x*.

→ **6 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (4, 2, 2.5, 'il')

These rectangles are similar. Find *x*.

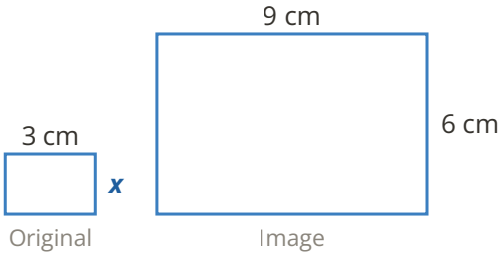
→ **10 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (3, 2, 3, 'ow')

These rectangles are similar. Find *x*.

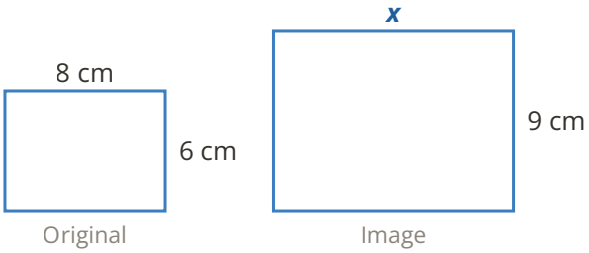
→ **2 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (8, 6, 1.5, 'il')

These rectangles are similar. Find *x*.

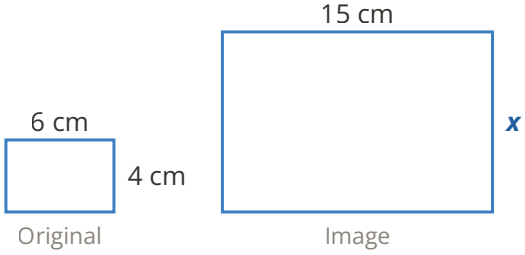
→ **12 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (6, 4, 2.5, 'iw')

These rectangles are similar. Find *x*.

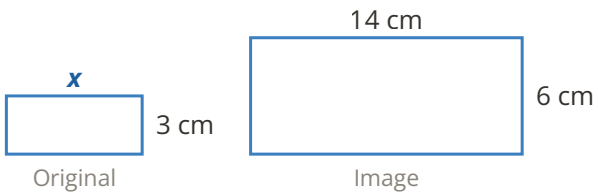
→ **10 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (7, 3, 2, 'ol')

These rectangles are similar. Find *x*.

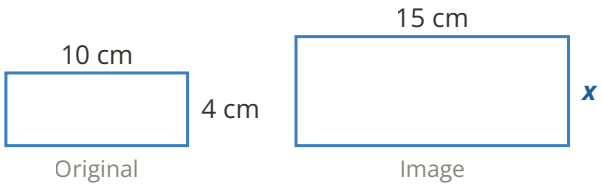
→ **7 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (10, 4, 1.5, 'iw')

These rectangles are similar. Find *x*.

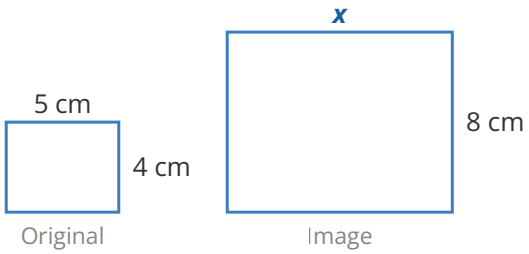
→ **6 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L1 · (5, 4, 2, 'il')

These rectangles are similar. Find *x*.

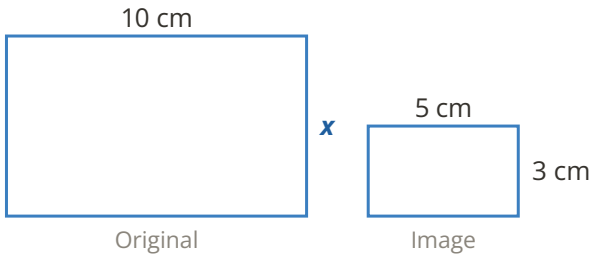
→ **10 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L2 · (10, 6, 0.5, 'ow')

These rectangles are similar. Find *x*.

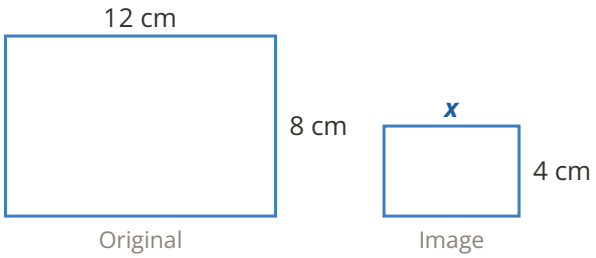
→ **6 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L2 · (12, 8, 0.5, 'il')

These rectangles are similar. Find *x*.

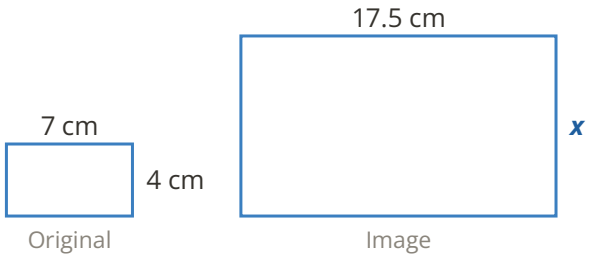
→ **6 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L2 · (7, 4, 2.5, 'iw')

These rectangles are similar. Find *x*.

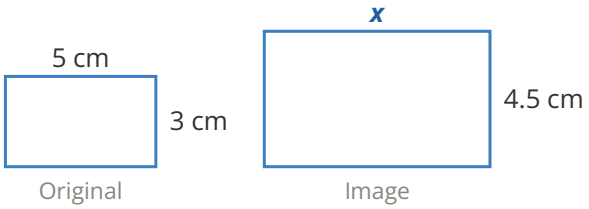
→ **10 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L2 · (5, 3, 1.5, 'il')

These rectangles are similar. Find *x*.

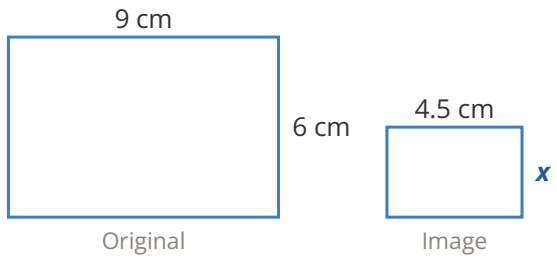
→ **7.5 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L2 · (9, 6, 0.5, 'iw')

These rectangles are similar. Find *x*.

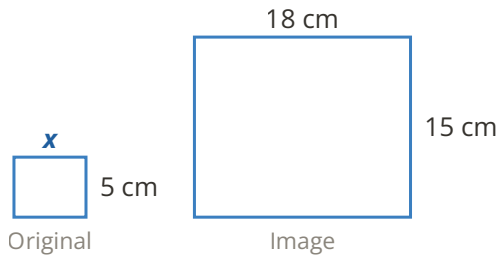
→ **3 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L2 · (6, 5, 3, 'ol')

These rectangles are similar. Find *x*.

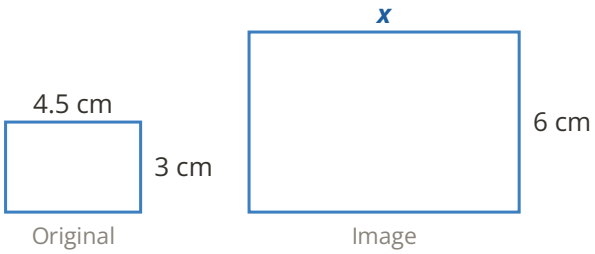
→ **6 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L2 · (4.5, 3, 2, 'il')

These rectangles are similar. Find *x*.

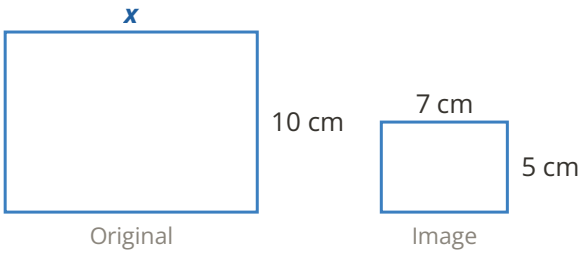
→ **9 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$



similar L2 · (14, 10, 0.5, 'ol')

These rectangles are similar. Find *x*.

→ **14 cm** Scale factor = $\frac{\text{image length}}{\text{original length}}$

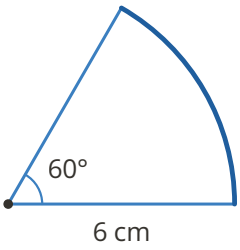


B AREA, SURFACE AREA AND VOLUME

arc L1 · (60, 6, 'cm')

Find the arc length, correct to 1 decimal place.

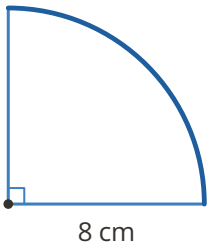
→ **6.3 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (90, 8, 'cm')

Find the arc length, correct to 1 decimal place.

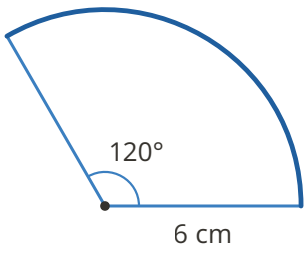
→ **12.6 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (120, 6, 'cm')

Find the arc length, correct to 1 decimal place.

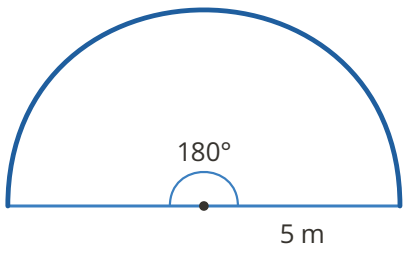
→ **12.6 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (180, 5, 'm')

Find the arc length, correct to 1 decimal place.

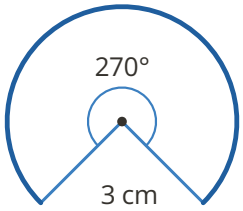
→ **15.7 m** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (270, 3, 'cm')

Find the arc length, correct to 1 decimal place.

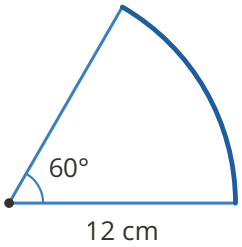
→ **14.1 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (60, 12, 'cm')

Find the arc length, correct to 1 decimal place.

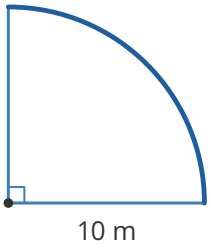
→ **12.6 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (90, 10, 'm')

Find the arc length, correct to 1 decimal place.

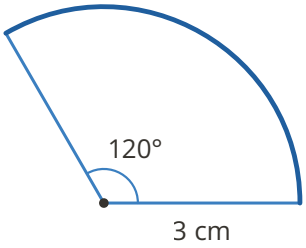
→ **15.7 m** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (120, 3, 'cm')

Find the arc length, correct to 1 decimal place.

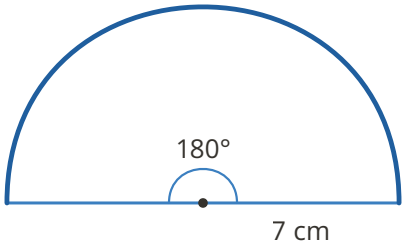
→ **6.3 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (180, 7, 'cm')

Find the arc length, correct to 1 decimal place.

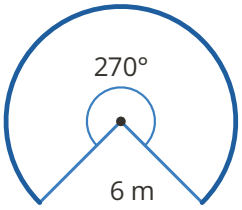
→ **22.0 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L1 · (270, 6, 'm')

Find the arc length, correct to 1 decimal place.

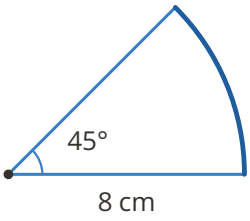
→ **28.3 m** $l = \frac{\theta}{360} \times 2\pi r$



arc L2 · (45, 8, 'cm')

Find the arc length, correct to 1 decimal place.

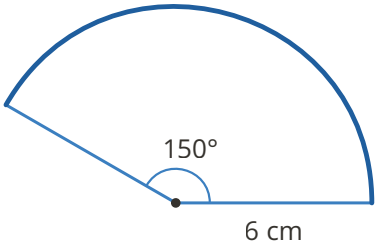
→ **6.3 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L2 · (150, 6, 'cm')

Find the arc length, correct to 1 decimal place.

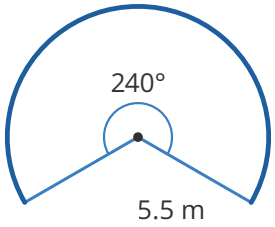
→ **15.7 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L2 · (240, 5.5, 'm')

Find the arc length, correct to 1 decimal place.

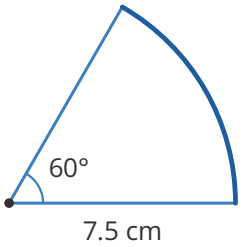
→ **23.0 m** $l = \frac{\theta}{360} \times 2\pi r$



arc L2 · (60, 7.5, 'cm')

Find the arc length, correct to 1 decimal place.

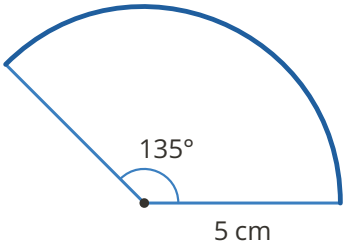
→ **7.9 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L2 · (135, 5, 'cm')

Find the arc length, correct to 1 decimal place.

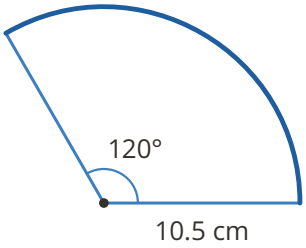
→ **11.8 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L2 · (120, 10.5, 'cm')

Find the arc length, correct to 1 decimal place.

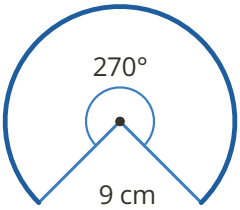
→ **22.0 cm** $l = \frac{\theta}{360} \times 2\pi r$



arc L2 · (270, 9, 'cm')

Find the arc length, correct to 1 decimal place.

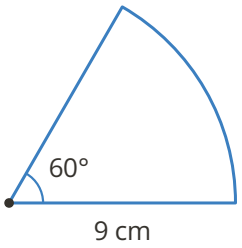
→ **42.4 cm** $l = \frac{\theta}{360} \times 2\pi r$



sector L1 · (60, 9, 'cm')

Find the area of this sector, correct to 1 decimal place.

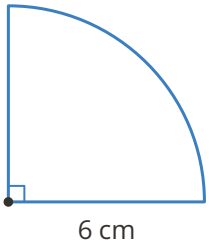
→ **42.4 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (90, 6, 'cm')

Find the area of this sector, correct to 1 decimal place.

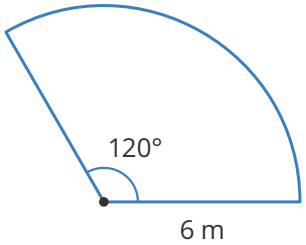
→ **28.3 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (120, 6, 'm')

Find the area of this sector, correct to 1 decimal place.

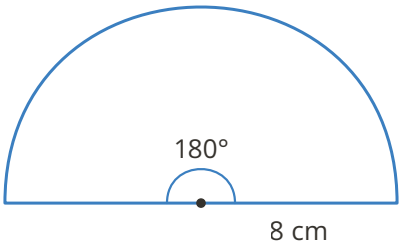
→ **37.7 m²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (180, 8, 'cm')

Find the area of this sector, correct to 1 decimal place.

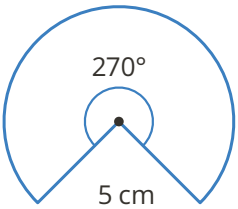
→ **100.5 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (270, 5, 'cm')

Find the area of this sector, correct to 1 decimal place.

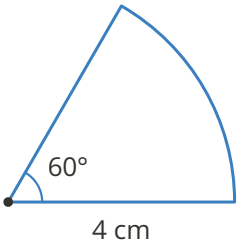
→ **58.9 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (60, 4, 'cm')

Find the area of this sector, correct to 1 decimal place.

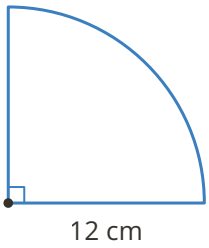
→ **8.4 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (90, 12, 'cm')

Find the area of this sector, correct to 1 decimal place.

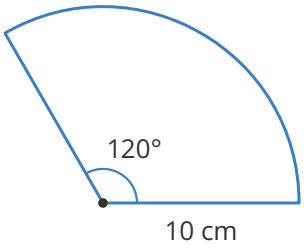
→ **113.1 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (120, 10, 'cm')

Find the area of this sector, correct to 1 decimal place.

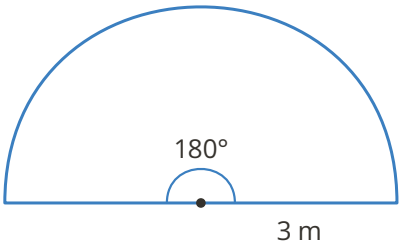
→ **104.7 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (180, 3, 'm')

Find the area of this sector, correct to 1 decimal place.

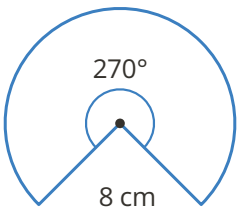
→ **14.1 m²** $A = \frac{\theta}{360} \times \pi r^2$



sector L1 · (270, 8, 'cm')

Find the area of this sector, correct to 1 decimal place.

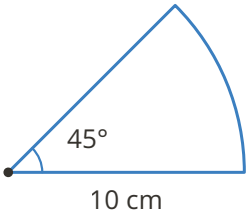
→ **150.8 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L2 · (45, 10, 'cm')

Find the area of this sector, correct to 1 decimal place.

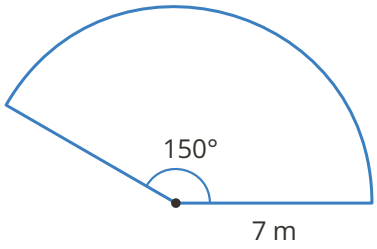
→ **39.3 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L2 · (150, 7, 'm')

Find the area of this sector, correct to 1 decimal place.

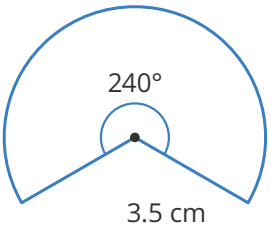
→ **64.1 m²** $A = \frac{\theta}{360} \times \pi r^2$



sector L2 · (240, 3.5, 'cm')

Find the area of this sector, correct to 1 decimal place.

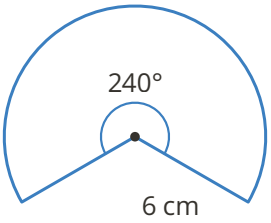
→ **25.7 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L2 · (240, 6, 'cm')

Find the area of this sector, correct to 1 decimal place.

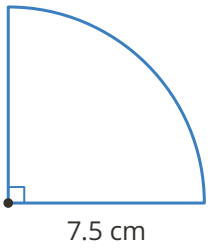
→ **75.4 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L2 · (90, 7.5, 'cm')

Find the area of this sector, correct to 1 decimal place.

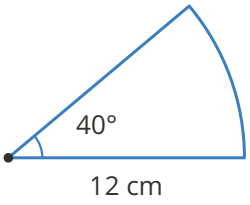
→ **44.2 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L2 · (40, 12, 'cm')

Find the area of this sector, correct to 1 decimal place.

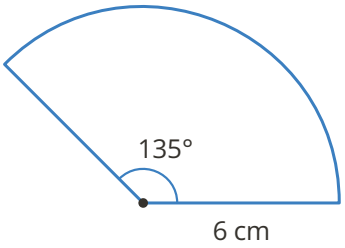
→ **50.3 cm²** $A = \frac{\theta}{360} \times \pi r^2$



sector L2 · (135, 6, 'cm')

Find the area of this sector, correct to 1 decimal place.

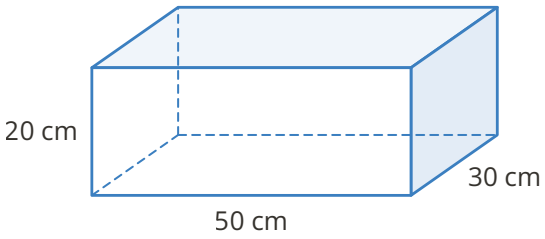
→ **42.4 cm²** $A = \frac{\theta}{360} \times \pi r^2$



capacity L1 · (50, 30, 20)

How many litres does this tank hold?

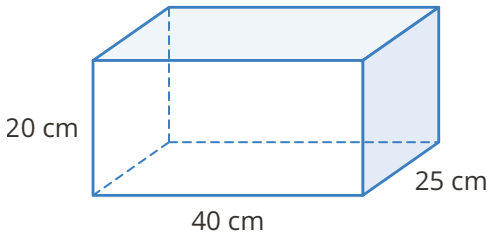
→ **30 L** $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$



capacity L1 · (40, 25, 20)

How many litres does this tank hold?

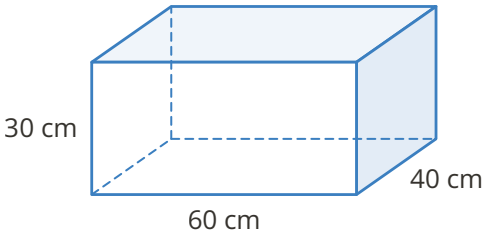
→ **20 L** $V = lwh$, $1000\ \text{cm}^3 = 1\ \text{L}$



capacity L1 · (60, 40, 30)

How many litres does this tank hold?

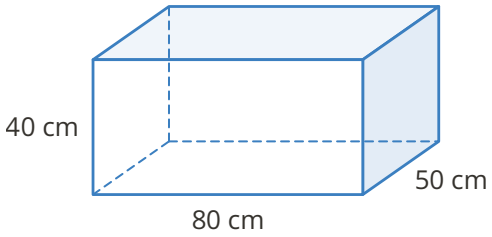
→ **72 L** $V = lwh$, 1000 cm³ = 1 L



capacity L1 · (80, 50, 40)

How many litres does this tank hold?

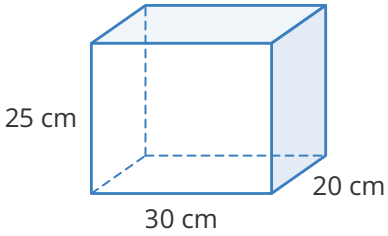
→ **160 L** $V = lwh$, 1000 cm³ = 1 L



capacity L1 · (30, 20, 25)

How many litres does this tank hold?

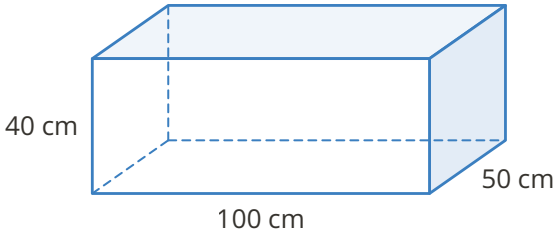
→ **15 L** $V = lwh$, 1000 cm³ = 1 L



capacity L1 · (100, 50, 40)

How many litres does this tank hold?

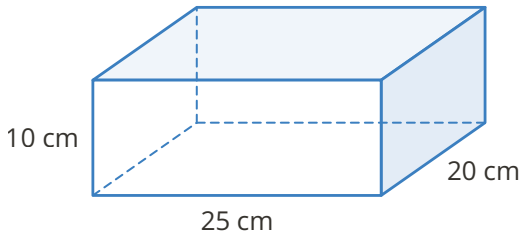
→ **200 L** $V = lwh$, 1000 cm³ = 1 L



capacity L1 · (25, 20, 10)

How many litres does this tank hold?

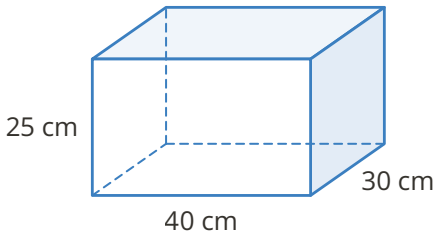
→ **5 L** $V = lwh$, 1000 cm³ = 1 L



capacity L1 · (40, 30, 25)

How many litres does this tank hold?

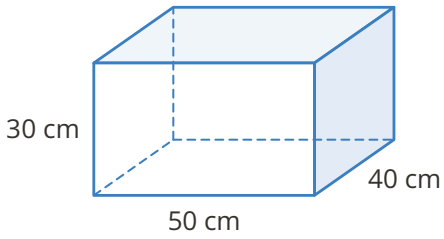
→ **30 L** $V = lwh$, 1000 cm³ = 1 L



capacity L1 · (50, 40, 30)

How many litres does this tank hold?

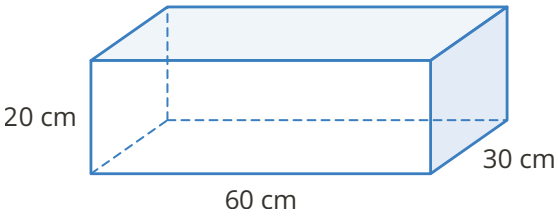
→ **60 L** $V = lwh$, 1000 cm³ = 1 L



capacity L1 · (60, 30, 20)

How many litres does this tank hold?

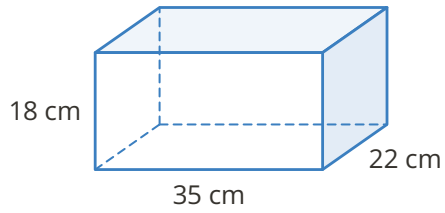
→ **36 L** $V = lwh$, 1000 cm³ = 1 L



capacity L2 · (35, 22, 18)

How many litres does this tank hold, correct to 1 decimal place?

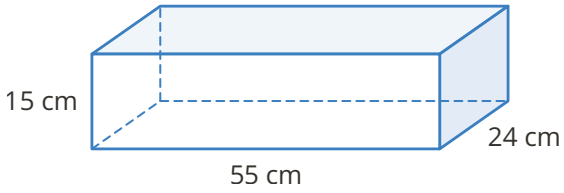
→ **13.9 L** $V = lwh$, 1000 cm³ = 1 L



capacity L2 · (55, 24, 15)

How many litres does this tank hold?

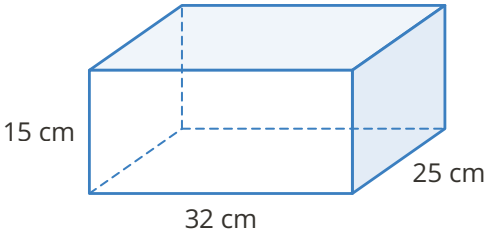
→ **19.8 L** $V = lwh$, 1000 cm³ = 1 L



capacity L2 · (32, 25, 15)

How many litres does this tank hold?

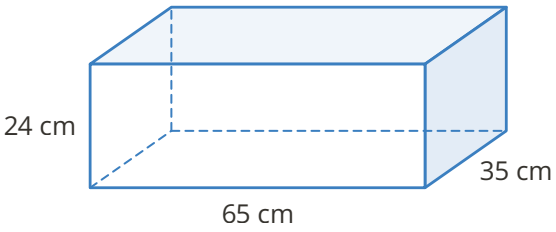
→ **12 L** $V = lwh$, 1000 cm³ = 1 L



capacity L2 · (65, 35, 24)

How many litres does this tank hold?

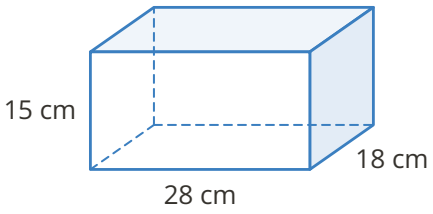
→ **54.6 L** $V = lwh$, 1000 cm³ = 1 L



capacity L2 · (28, 18, 15)

How many litres does this tank hold, correct to 1 decimal place?

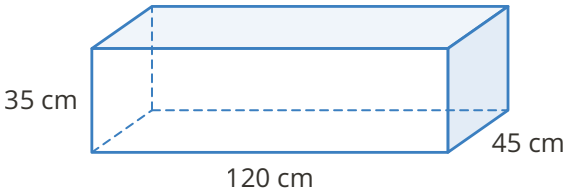
→ **7.6 L** $V = lwh$, 1000 cm³ = 1 L



capacity L2 · (120, 45, 35)

How many litres does this tank hold?

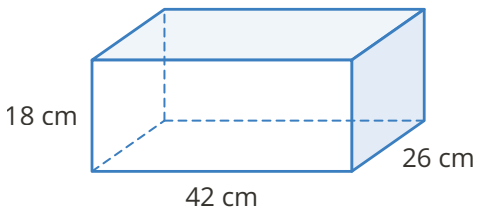
→ **189 L** $V = lwh$, 1000 cm³ = 1 L



capacity L2 · (42, 26, 18)

How many litres does this tank hold, correct to 1 decimal place?

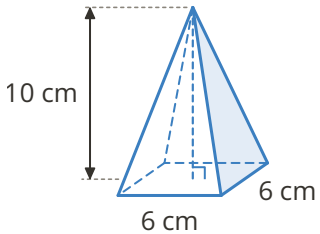
→ **19.7 L** $V = lwh$, 1000 cm³ = 1 L



v_pyramid L1 · T4 only, formula printed · (6, 6, 10, 'cm')

Find the volume of this pyramid.

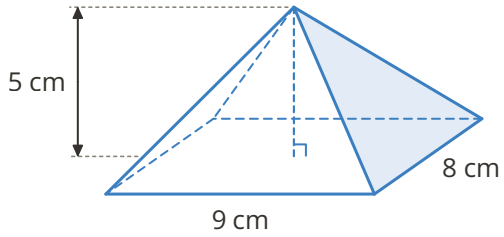
→ **120 cm³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L1 · T4 only, formula printed · (9, 8, 5, 'cm')

Find the volume of this pyramid.

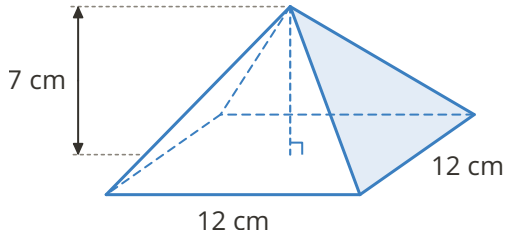
→ **120 cm³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L1 · T4 only, formula printed · (12, 12, 7, 'cm')

Find the volume of this pyramid.

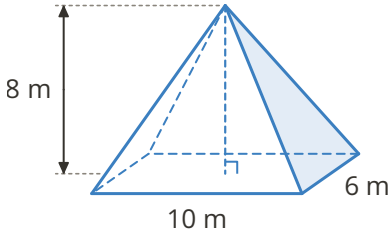
→ **336 cm³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L1 · T4 only, formula printed · (10, 6, 8, 'm')

Find the volume of this pyramid.

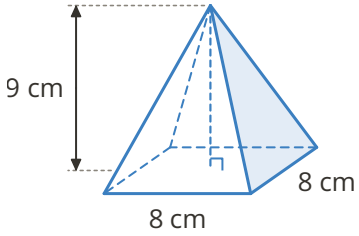
→ **160 m³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L1 · T4 only, formula printed · (8, 8, 9, 'cm')

Find the volume of this pyramid.

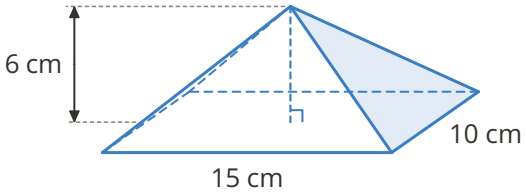
→ **192 cm³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L1 · T4 only, formula printed · (15, 10, 6, 'cm')

Find the volume of this pyramid.

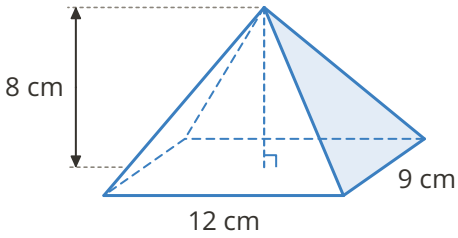
→ **300 cm³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L2 · T4 only, formula printed · (12, 9, 8, 'cm')

Find the volume of this pyramid.

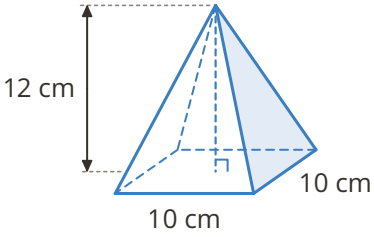
→ **288 cm³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L2 · T4 only, formula printed · (10, 10, 12, 'cm')

Find the volume of this pyramid.

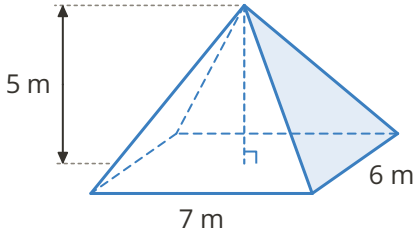
→ **400 cm³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L2 · T4 only, formula printed · (7, 6, 5, 'm')

Find the volume of this pyramid.

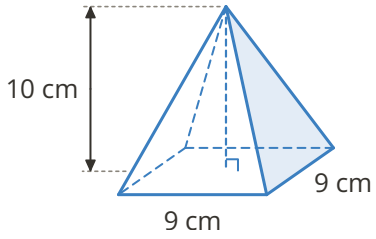
→ **70 m³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_pyramid L2 · T4 only, formula printed · (9, 9, 10, 'cm')

Find the volume of this pyramid.

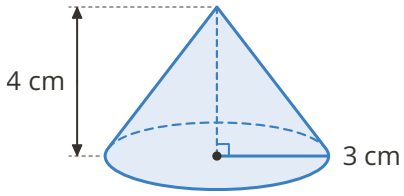
→ **270 cm³** $V = \frac{1}{3}Ah$ (A = area of the base)



v_cone L1 · T4 only, formula printed · (3, 4, False, 'cm')

Find the volume of this cone, correct to 1 decimal place.

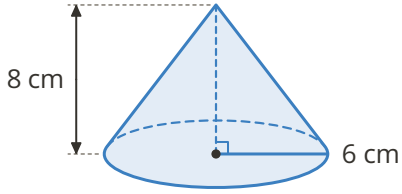
→ **37.7 cm³** $V = \frac{1}{3}\pi r^2 h$



v_cone L1 · T4 only, formula printed · (6, 8, False, 'cm')

Find the volume of this cone, correct to 1 decimal place.

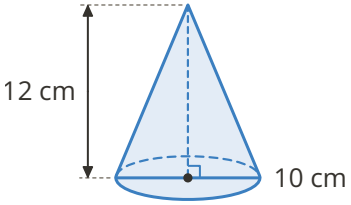
→ **301.6 cm³** $V = \frac{1}{3}\pi r^2 h$



v_cone L1 · T4 only, formula printed · (5, 12, True, 'cm')

Find the volume of this cone, correct to 1 decimal place.

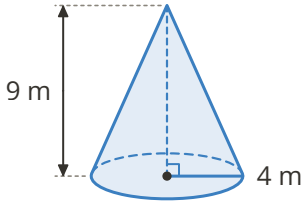
→ **314.2 cm³** $V = \frac{1}{3}\pi r^2 h$



v_cone L1 · T4 only, formula printed · (4, 9, False, 'm')

Find the volume of this cone, correct to 1 decimal place.

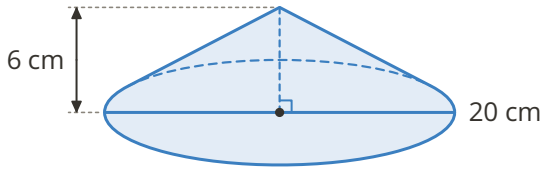
→ **150.8 m³** $V = \frac{1}{3}\pi r^2 h$



v_cone L1 · T4 only, formula printed · (10, 6, True, 'cm')

Find the volume of this cone, correct to 1 decimal place.

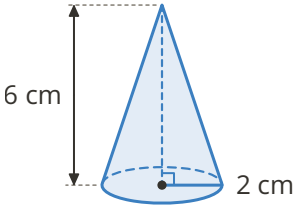
→ **628.3 cm³** $V = \frac{1}{3}\pi r^2 h$



v_cone L1 · T4 only, formula printed · (2, 6, False, 'cm')

Find the volume of this cone, correct to 1 decimal place.

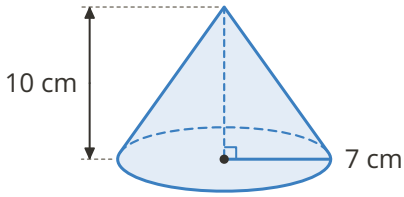
→ **25.1 cm³** $V = \frac{1}{3}\pi r^2 h$



v_cone L2 · T4 only, formula printed · (7, 10, False, 'cm')

Find the volume of this cone, correct to 1 decimal place.

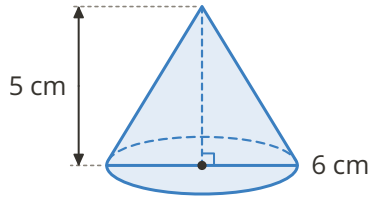
→ **513.1 cm³** $V = \frac{1}{3}\pi r^2 h$



v_cone L2 · T4 only, formula printed · (3, 5, True, 'cm')

Find the volume of this cone, correct to 1 decimal place.

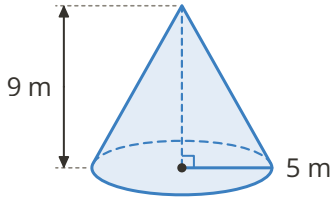
→ **47.1 cm³** $V = \frac{1}{3}\pi r^2 h$



v_cone L2 · T4 only, formula printed · (5, 9, False, 'm')

Find the volume of this cone, correct to 1 decimal place.

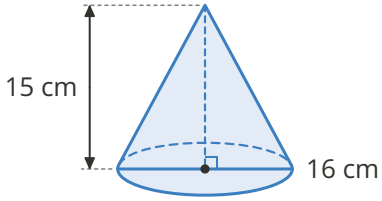
→ **235.6 m³** $V = \frac{1}{3}\pi r^2 h$



v_cone L2 · T4 only, formula printed · (8, 15, True, 'cm')

Find the volume of this cone, correct to 1 decimal place.

→ **1005.3 cm³** $V = \frac{1}{3}\pi r^2 h$



C PERCENTAGES AND MONEY

profit_pct L1 · (40, 50)

Find the percentage profit.

→ **25% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$40
Selling price	$50

profit_pct L1 · (60, 78)

Find the percentage profit.

→ **30% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$60
Selling price	$78

profit_pct L1 · (50, 60)

Find the percentage profit.

→ **20% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$50
Selling price	$60

profit_pct L1 · (200, 260)

Find the percentage profit.

→ **30% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$200
Selling price	$260

profit_pct L1 · (75, 60)

Find the percentage loss.

→ **20% loss** Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$

Cost price	$75
Selling price	$60

profit_pct L1 · (80, 68)

Find the percentage loss.

→ **15% loss** Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$

Cost price	$80
Selling price	$68

profit_pct L1 · (30, 45)

Find the percentage profit.

→ **50% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$30
Selling price	$45

profit_pct L1 · (250, 300)

Find the percentage profit.

→ **20% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$250
Selling price	$300

profit_pct L1 · (120, 90)

Find the percentage loss.

→ **25% loss** Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$

Cost price	$120
Selling price	$90

profit_pct L1 · (160, 200)

Find the percentage profit.

→ **25% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$160
Selling price	$200

profit_pct L2 · (12.5, 15)

Find the percentage profit.

→ **20% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$12.50
Selling price	$15

profit_pct L2 · (24.8, 31)

Find the percentage profit.

→ **25% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$24.80
Selling price	$31

profit_pct L2 · (45, 36)

Find the percentage loss.

→ **20% loss** Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$

Cost price	$45
Selling price	$36

profit_pct L2 · (2.5, 3.1)

Find the percentage profit.

→ **24% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$2.50
Selling price	$3.10

profit_pct L2 · (850, 1020)

Find the percentage profit.

→ **20% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$850
Selling price	$1020

profit_pct L2 · (125, 115)

Find the percentage loss.

→ **8% loss** Percentage loss = $\frac{\text{loss}}{\text{cost price}} \times 100\%$

Cost price	$125
Selling price	$115

profit_pct L2 · (36.4, 45.5)

Find the percentage profit.

→ **25% profit** Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

Cost price	$36.40
Selling price	$45.50

interest L1 · (2000, 4, 3, '3 years', 'inv')

Find the simple interest.

→

$240 $I = Prn$ (interest = amount × yearly interest rate × number of years)

Amount invested	$2000
Interest rate	4% per year
Time	3 years

interest L1 · (5000, 3, 2, '2 years', 'inv')

Find the simple interest.

→

$300 $I = Prn$ (interest = amount × yearly interest rate × number of years)

Amount invested	$5000
Interest rate	3% per year
Time	2 years

interest L1 · (1500, 6, 4, '4 years', 'inv')

Find the simple interest.

→

$360 $I = Prn$ (interest = amount × yearly interest rate × number of years)

Amount invested	$1500
Interest rate	6% per year
Time	4 years

interest L1 · (800, 5, 3, '3 years', 'inv')

Find the simple interest.

→

$120 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$800
Interest rate	5% per year
Time	3 years

interest L1 · (12000, 2, 5, '5 years', 'inv')

Find the simple interest.

→

$1200 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$12 000
Interest rate	2% per year
Time	5 years

interest L1 · (4000, 8, 2, '2 years', 'loan')

Find the simple interest on this loan.

→

$640 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount borrowed	$4000
Interest rate	8% per year
Time	2 years

interest L1 · (2500, 4, 2, '2 years', 'inv')

Find the simple interest.

→

$200 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$2500
Interest rate	4% per year
Time	2 years

interest L1 · (3000, 7, 3, '3 years', 'loan')

Find the simple interest on this loan.

→

$630 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount borrowed	$3000
Interest rate	7% per year
Time	3 years

interest L1 · (600, 5, 2, '2 years', 'inv')

Find the simple interest.

→ **$60** *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$600
Interest rate	5% per year
Time	2 years

interest L2 · (2000, 3.5, Fraction(5, 2), '2½ years', 'inv')

Find the simple interest.

→

$175 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$2000
Interest rate	3.5% per year
Time	2½ years

interest L2 · (6000, 4, Fraction(3, 2), '18 months', 'loan')

Find the simple interest on this loan.

→

$360 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount borrowed	$6000
Interest rate	4% per year
Time	18 months

interest L2 · (4500, 5, Fraction(1, 2), '6 months', 'inv')

Find the simple interest.

→

$112.50 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$4500
Interest rate	5% per year
Time	6 months

interest L2 · (1200, 4.5, 3, '3 years', 'inv')

Find the simple interest.

→

$162 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$1200
Interest rate	4.5% per year
Time	3 years

interest L2 · (8000, 6, Fraction(3, 4), '9 months', 'loan')

Find the simple interest on this loan.

→

$360 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount borrowed	$8000
Interest rate	6% per year
Time	9 months

interest L2 · (3500, 2.8, Fraction(9, 2), '4½ years', 'inv')

Find the simple interest.

→

$441 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$3500
Interest rate	2.8% per year
Time	4½ years

interest L2 · (15000, 5.5, 2, '2 years', 'inv')

Find the simple interest.

→

$1650 *I = Prn* (interest = amount × yearly interest rate × number of years)

Amount invested	$15 000
Interest rate	5.5% per year
Time	2 years

wages L1 · ('Ava', 25, 30, 0, None)

Ava earns $25 an hour and works 30 hours. How much does Ava earn?

→ **$750** Pay = hourly rate × hours

wages L1 · ('Noah', 22, 20, 0, None)

Noah earns $22 an hour and works 20 hours. How much does Noah earn?

→ **$440** Pay = hourly rate × hours

wages L1 · ('Mia', 30, 15, 0, None)

Mia earns $30 an hour and works 15 hours. How much does Mia earn?

→ **$450** Pay = hourly rate × hours

wages L1 · ('Leo', 28, 35, 0, None)

Leo earns $28 an hour and works 35 hours. How much does Leo earn?

→ **$980** Pay = hourly rate × hours

wages L1 · ('Liam', 24, 38, 4, 't15')

Find Liam’s total pay for the week.

→

$1056 Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate

Hourly rate	$24
Normal hours	38 hours
Time-and-a-half	4 hours

wages L1 · ('Zara', 30, 35, 3, 'dt')

Find Zara’s total pay for the week.

→ **$1230** Pay = hourly rate × hours; double time = 2 × hourly rate

Hourly rate	$30
Normal hours	35 hours
Double time	3 hours

wages L1 · ('Kai', 26, 38, 6, 't15')

Find Kai’s total pay for the week.

→

$1222 Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate

Hourly rate	$26
Normal hours	38 hours
Time-and-a-half	6 hours

wages L1 · ('Ruby', 32, 30, 2, 'dt')

Find Ruby’s total pay for the week.

→ **$1088** Pay = hourly rate × hours; double time = 2 × hourly rate

Hourly rate	$32
Normal hours	30 hours
Double time	2 hours

wages L1 · ('Jack', 28, 36, 5, 't15')

Find Jack’s total pay for the week.

→

$1218 Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate

Hourly rate	$28
Normal hours	36 hours
Time-and-a-half	5 hours

wages L1 · ('Priya', 20, 38, 4, 'dt')

Find Priya’s total pay for the week.

→ **$920** Pay = hourly rate × hours; double time = 2 × hourly rate

Hourly rate	$20
Normal hours	38 hours
Double time	4 hours

wages L2 · ('Ella', 25.5, 38, 4, 't15')

Find Ella’s total pay for the week.

→

$1122 Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate

Hourly rate	$25.50
Normal hours	38 hours
Time-and-a-half	4 hours

wages L2 · ('Tom', 31.4, 35, 6, 'dt')

Find Tom’s total pay for the week.

→ **$1475.80** Pay = hourly rate × hours; double time = 2 × hourly rate

Hourly rate	$31.40
Normal hours	35 hours
Double time	6 hours

wages L2 · ('Aisha', 27.8, 32, 3, 't15')

Find Aisha’s total pay for the week.

→ **$1014.70** Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate

Hourly rate	$27.80
Normal hours	32 hours
Time-and-a-half	3 hours

wages L2 · ('Hamish', 23.6, 38, 0, None)

Hamish earns $23.60 an hour and works 38 hours. How much does Hamish earn?

→ **$896.80** Pay = hourly rate × hours

wages L2 · ('Sienna', 29.9, 36, 3, 't15')

Find Sienna’s total pay for the week.

→ **$1210.95** Pay = hourly rate × hours; time-and-a-half = 1.5 × hourly rate

Hourly rate	$29.90
Normal hours	36 hours
Time-and-a-half	3 hours

wages L2 · ('Arjun', 26.75, 30, 4, 'dt')

Find Arjun’s total pay for the week.

→ **$1016.50** Pay = hourly rate × hours; double time = 2 × hourly rate

Hourly rate	$26.75
Normal hours	30 hours
Double time	4 hours

salary L1 · ('Liam', 62400, 'w')

Liam earns a salary of $62 400 a year. Find Liam’s weekly pay.

→ **$1200** Weekly pay = yearly salary ÷ 52

salary L1 · ('Ava', 52000, 'f')

Ava earns a salary of $52 000 a year. Find Ava’s fortnightly pay.

→ **$2000** Fortnightly pay = yearly salary ÷ 26

salary L1 · ('Mia', 78000, 'w')

Mia earns a salary of $78 000 a year. Find Mia’s weekly pay.

→ **$1500** Weekly pay = yearly salary ÷ 52

salary L1 · ('Noah', 57200, 'f')

Noah earns a salary of $57 200 a year. Find Noah's fortnightly pay.

→ **$2200** Fortnightly pay = yearly salary ÷ 26

salary L1 · ('Zara', 67600, 'w')

Zara earns a salary of $67 600 a year. Find Zara's weekly pay.

→ **$1300** Weekly pay = yearly salary ÷ 52

salary L1 · ('Kai', 83200, 'f')

Kai earns a salary of $83 200 a year. Find Kai's fortnightly pay.

→ **$3200** Fortnightly pay = yearly salary ÷ 26

salary L1 · ('Ruby', 91000, 'w')

Ruby earns a salary of $91 000 a year. Find Ruby's weekly pay.

→ **$1750** Weekly pay = yearly salary ÷ 52

salary L1 · ('Jack', 54600, 'f')

Jack earns a salary of $54 600 a year. Find Jack's fortnightly pay.

→ **$2100** Fortnightly pay = yearly salary ÷ 26

salary L2 · ('Ella', 68500, 'w')

Ella earns a salary of $68 500 a year. Find Ella's weekly pay, to the nearest cent.

→ **$1317.31** Weekly pay = yearly salary ÷ 52

salary L2 · ('Tom', 74250, 'f')

Tom earns a salary of $74 250 a year. Find Tom's fortnightly pay, to the nearest cent.

→ **$2855.77** Fortnightly pay = yearly salary ÷ 26

salary L2 · ('Aisha', 82750, 'w')

Aisha earns a salary of $82 750 a year. Find Aisha's weekly pay, to the nearest cent.

→ **$1591.35** Weekly pay = yearly salary ÷ 52

salary L2 · ('Leo', 95300, 'f')

Leo earns a salary of $95 300 a year. Find Leo's fortnightly pay, to the nearest cent.

→ **$3665.38** Fortnightly pay = yearly salary ÷ 26

salary L2 · ('Priya', 61250, 'w')

Priya earns a salary of $61 250 a year. Find Priya's weekly pay, to the nearest cent.

→ **$1177.88** Weekly pay = yearly salary ÷ 52

salary L2 · ('Hamish', 58900, 'f')

Hamish earns a salary of $58 900 a year. Find Hamish's fortnightly pay, to the nearest cent.

→ **$2265.38** Fortnightly pay = yearly salary ÷ 26

commission L1 · ('Zara', 5, 8000)

Find Zara's commission.

→ **$400** Commission = percentage × value of sales

Value of sales	$8000
Commission rate	5%

commission L1 · ('Ethan', 2, 650000)

Find Ethan's commission.

→ **$13 000** Commission = percentage × value of sales

Value of sales	$650 000
Commission rate	2%

commission L1 · ('Grace', 10, 4500)

Find Grace's commission.

→ **$450** Commission = percentage × value of sales

Value of sales	$4500
Commission rate	10%

commission L1 · ('Arjun', 3, 120000)

Find Arjun's commission.

→ **$3600** Commission = percentage × value of sales

Value of sales	$120 000
Commission rate	3%

commission L1 · ('Chloe', 4, 25000)

Find Chloe's commission.

→ **$1000** Commission = percentage × value of sales

Value of sales	$25 000
Commission rate	4%

commission L1 · ('Mason', 15, 2400)

Find Mason's commission.

→ **$360** Commission = percentage × value of sales

Value of sales	$2400
Commission rate	15%

commission L1 · ('Isla', 6, 9500)

Find Isla's commission.

→ **$570** Commission = percentage × value of sales

Value of sales	$9500
Commission rate	6%

commission L1 · ('Kai', 2, 480000)

Find Kai's commission.

→ **$9600** Commission = percentage × value of sales

Value of sales	$480 000
Commission rate	2%

commission L2 · ('base', 'Noah', 450, 3, 12500)

Find Noah's total pay for the week.

→ **$825** Pay = base wage + percentage × value of sales

Weekly base wage	$450
Commission	3% of sales
Sales this week	$12 500

commission L2 · ('base', 'Ruby', 380, 5, 7860)

Find Ruby's total pay for the week.

→ **$773** Pay = base wage + percentage × value of sales

Weekly base wage	$380
Commission	5% of sales
Sales this week	$7860

commission L2 · ('base', 'Leo', 520, 2.5, 18400)

Find Leo's total pay for the week.

→ **$980** Pay = base wage + percentage × value of sales

Weekly base wage	$520
Commission	2.5% of sales
Sales this week	$18 400

commission L2 · ('base', 'Mia', 600, 4, 9375)

Find Mia's total pay for the week.

→ **$975** Pay = base wage + percentage × value of sales

Weekly base wage	$600
Commission	4% of sales
Sales this week	$9375

commission L2 · ('Tom', 2.5, 685000)

Find Tom's commission.

→ **$17 125** Commission = percentage × value of sales

Value of sales	$685 000
Commission rate	2.5%

commission L2 · ('Ava', 1.8, 540000)

Find Ava’s commission.

→ **$9720** Commission = percentage × value of sales

Value of sales	$540 000
Commission rate	1.8%

holiday_pay L1 · ('Jo', 1200)

Find Jo’s total holiday pay.

→ **$5640** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1200
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L1 · ('Sam', 950)

Find Sam’s total holiday pay.

→ **$4465** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$950
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L1 · ('Ali', 1100)

Find Ali’s total holiday pay.

→ **$5170** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1100
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L1 · ('Bella', 800)

Find Bella’s total holiday pay.

→ **$3760** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$800
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L1 · ('Finn', 1350)

Find Finn’s total holiday pay.

→ **$6345** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1350
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L1 · ('Tara', 1040)

Find Tara’s total holiday pay.

→ **$4888** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1040
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L1 · ('Omar', 1500)

Find Omar’s total holiday pay.

→ **$7050** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1500
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L1 · ('Lucy', 870)

Find Lucy’s total holiday pay.

→ **$4089** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$870
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L2 · ('Eva', 1145)

Find Eva’s total holiday pay.

→ **$5381.50** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1145
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L2 · ('Ben', 985)

Find Ben’s total holiday pay.

→ **$4629.50** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$985
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L2 · ('Nina', 1263.5)

Find Nina’s total holiday pay.

→ **$5938.45** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1263.50
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L2 · ('Raj', 1427)

Find Raj’s total holiday pay.

→ **$6706.90** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1427
Holiday	4 weeks
Leave loading	17.5%

holiday_pay L2 · ('Ivy', 1098.4)

Find Ivy’s total holiday pay.

→ **$5162.48** Holiday pay = 4 weeks’ pay + 17.5% of 4 weeks’ pay

Weekly pay	$1098.40
Holiday	4 weeks
Leave loading	17.5%

zero_neg L1 · ('neg', 2, 3)

Evaluate.

→ **1/8** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

2^{-3}

zero_neg L1 · ('zplus', 7, 3)

Evaluate.

→ **4** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

$7^0 + 3$

zero_neg L1 · ('pos', 'x', 2)

Write with a positive index.

→ **1/x^2** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

x^{-2}

zero_neg L1 · ('neg', 10, 2)

Evaluate.

→ **1/100** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

10^{-2}

zero_neg L1 · ('zero', 9)

Evaluate.

→ **1** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

9^0

zero_neg L1 · ('neg', 5, 1)

Evaluate.

→ **1/5** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

5^{-1}

zero_neg L1 · ('zcoef', 6, 'a')

Simplify.

→ **6** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

$6a^0$

zero_neg L1 · ('neg', 3, 2)

Evaluate.

→ **1/9** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

3^{-2}

zero_neg L1 · ('pos', 'm', 4)

Write with a positive index.

→ **1/m^4** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

m^{-4}

zero_neg L1 · ('zsum', 5, 2)

Evaluate.

→ **2** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

$5^0 + 2^0$

zero_neg L2 · ('poscoef', 3, 'x', 2)

Write with a positive index.

→ **3/x^2** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

$3x^{-2}$

zero_neg L2 · ('neg', 4, 2)

Evaluate.

→ **1/16** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

4^{-2}

zero_neg L2 · ('negsum', 2)

Evaluate.

→ **3/2** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

$2^{-1} + 2^0$

zero_neg L2 · ('zbr', 5, 'y')

Simplify.

→ **1** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

$(5y)^0$

zero_neg L2 · ('neg', 10, 3)

Evaluate.

→ **1/1000** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

10^{-3}

zero_neg L2 · ('pos', 'a', 5)

Write with a positive index.

→ **1/a^5** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

a^{-5}

zero_neg L2 · ('poscoef', 5, 'k', 3)

Write with a positive index.

→ **5/k^3** $a^0 = 1, \quad a^{-n} = \frac{1}{a^n}$

$5k^{-3}$

sci L1 · ('to_sci', '45000')

Write in scientific notation.

→ **4.5 × 10^4** $a \times 10^n$, where $1 \le a < 10$

45 000

sci L1 · ('to_sci', '3600000')

Write in scientific notation.

→ **3.6 × 10^6** $a \times 10^n$, where $1 \le a < 10$

3 600 000

sci L1 · ('to_sci', '820')

Write in scientific notation.

→ **8.2 × 10^2** $a \times 10^n$, where $1 \le a < 10$

820

sci L1 · ('to_sci', '7000')

Write in scientific notation.

→ **7 × 10^3** $a \times 10^n$, where $1 \le a < 10$

7000

sci L1 · ('to_sci', '150000')

Write in scientific notation.

→ **1.5 × 10^5** $a \times 10^n$, where $1 \le a < 10$

150 000

sci L1 · ('to_num', '3.2', 4)

Write as a basic numeral.

→ **32 000** $a \times 10^n$, where $1 \le a < 10$

3.2×10^4

sci L1 · ('to_num', '5', 3)

Write as a basic numeral.

→ **5000** $a \times 10^n$, where $1 \le a < 10$

5×10^3

sci L1 · ('to_num', '6.1', 5)

Write as a basic numeral.

→ **610 000** $a \times 10^n$, where $1 \le a < 10$

6.1×10^{5}

sci L1 · ('to_num', '2.75', 6)

Write as a basic numeral.

→ **2 750 000** $a \times 10^n$, where $1 \le a < 10$

2.75×10^{6}

sci L1 · ('to_num', '9.4', 2)

Write as a basic numeral.

→ **940** $a \times 10^n$, where $1 \le a < 10$

9.4×10^{2}

sci L2 · ('to_sci', '0.0062')

Write in scientific notation.

→ **6.2 × 10^−3** $a \times 10^n$, where $1 \le a < 10$

0.0062

sci L2 · ('to_sci', '0.00045')

Write in scientific notation.

→ **4.5 × 10^−4** $a \times 10^n$, where $1 \le a < 10$

0.00045

sci L2 · ('to_num', '7.5', -4)

Write as a basic numeral.

→ **0.00075** $a \times 10^n$, where $1 \le a < 10$

7.5×10^{-4}

sci L2 · ('to_num', '3.08', -2)

Write as a basic numeral.

→ **0.0308** $a \times 10^n$, where $1 \le a < 10$

3.08×10^{-2}

sci L2 · ('to_sci', '28500000')

Write in scientific notation.

→ **2.85 × 10^7** $a \times 10^n$, where $1 \le a < 10$

28 500 000

sci L2 · ('to_sci', '0.9')

Write in scientific notation.

→ **9 × 10^−1** $a \times 10^n$, where $1 \le a < 10$

0.9

sci L2 · ('to_num', '1.6', -5)

Write as a basic numeral.

→ **0.000016** $a \times 10^n$, where $1 \le a < 10$

1.6×10^{-5}

formula_sub L1 · ('fs_v', 'v', {'u': 5, 'a': 3, 't': 4})

Find v when $u = 5$, $a = 3$ and $t = 4$.

→ **17** $v = u + at$

$v = u + at$

formula_sub L1 · ('fs_v', 't', {'v': 26, 'u': 6, 'a': 4})

Find t when $v = 26$, $u = 6$ and $a = 4$.

→ **5** $v = u + at$

$v = u + at$

formula_sub L1 · ('fs_P', 'P', {'l': 8, 'w': 5})

Find P when $l = 8$ and $w = 5$.

→ **26** $P = 2(l + w)$

$P = 2(l + w)$

formula_sub L1 · ('fs_P', 'w', {'P': 30, 'l': 9})

Find w when $P = 30$ and $l = 9$.

→ **6** $P = 2(l + w)$

$P = 2(l + w)$

formula_sub L1 · ('fs_C', 'C', {'d': 10})

Find C when $d = 10$.

→ **29** $C = 5 + 2.4d$

$C = 5 + 2.4d$

formula_sub L1 · ('fs_C', 'd', {'C': 29})

Find d when $C = 29$.

→ **10** $C = 5 + 2.4d$

$C = 5 + 2.4d$

formula_sub L1 · ('fs_F', 'F', {'C': 25})

Find F when $C = 25$.

→ **77** $F = 1.8C + 32$

$$F = 1.8C + 32$$

formula_sub L1 · ('fs_F', 'C', {'F': 50})

Find C when $F = 50$.

→ **10** $F = 1.8C + 32$

$$F = 1.8C + 32$$

formula_sub L1 · ('fs_A', 'h', {'A': 24, 'b': 8})

Find h when $A = 24$ and $b = 8$.

→ **6** $A = ½bh$

$$A = ½bh$$

formula_sub L1 · ('fs_A', 'h', {'A': 35, 'b': 10})

Find h when $A = 35$ and $b = 10$.

→ **7** $A = ½bh$

$$A = ½bh$$

formula_sub L1 · ('fs_y', 'x', {'y': 17, 'm': 3, 'c': 2})

Find x when $y = 17$, $m = 3$ and $c = 2$.

→ **5** $y = mx + c$

$$y = mx + c$$

formula_sub L1 · ('fs_v', 'a', {'v': 30, 'u': 10, 't': 5})

Find a when $v = 30$, $u = 10$ and $t = 5$.

→ **4** $v = u + at$

$$v = u + at$$

formula_sub L2 · ('fs_v', 'v', {'u': 2.5, 'a': 1.5, 't': 6})

Find v when $u = 2.5$, $a = 1.5$ and $t = 6$.

→ **11.5** $v = u + at$

$$v = u + at$$

formula_sub L2 · ('fs_v', 't', {'v': 20, 'u': 4, 'a': 2.5})

Find t when $v = 20$, $u = 4$ and $a = 2.5$.

→ **6.4** $v = u + at$

$$v = u + at$$

formula_sub L2 · ('fs_P', 'l', {'P': 37, 'w': 6.5})

Find l when $P = 37$ and $w = 6.5$.

→ **12** $P = 2(l + w)$

$$P = 2(l + w)$$

formula_sub L2 · ('fs_C', 'd', {'C': 17})

Find d when $C = 17$.

→ **5** $C = 5 + 2.4d$

$$C = 5 + 2.4d$$

formula_sub L2 · ('fs_F', 'C', {'F': 95})

Find C when $F = 95$.

→ **35** $F = 1.8C + 32$

$$F = 1.8C + 32$$

formula_sub L2 · ('fs_F', 'F', {'C': -5})

Find F when $C = -5$.

→ **23** $F = 1.8C + 32$

$$F = 1.8C + 32$$

formula_sub L2 · ('fs_A', 'h', {'A': 27, 'b': 12})

Find h when $A = 27$ and $b = 12$.

→ **4.5** $A = ½bh$

$$A = ½bh$$

formula_sub L2 · ('fs_y', 'x', {'y': -7, 'm': 2, 'c': 5})

Find x when $y = -7$, $m = 2$ and $c = 5$.

→ **−6** $y = mx + c$

$$y = mx + c$$

formula_sub L2 · ('fs_y', 'x', {'y': 11, 'm': 4, 'c': -3})

Find x when $y = 11$, $m = 4$ and $c = -3$.

→ **3.5** $y = mx + c$

$$y = mx + c$$

ratio L1 · (120, (3, 5), '$')

Share $120 in the ratio 3 : 5. Give both shares.

→ **$45 and $75** One part = amount ÷ total number of parts

ratio L1 · (84, (4, 3), 'lollies')

Share 84 lollies in the ratio 4 : 3. Give both shares.

→ **48 and 36** One part = amount ÷ total number of parts

ratio L1 · (200, (1, 4), '$')

Share $200 in the ratio 1 : 4. Give both shares.

→ **$40 and $160** One part = amount ÷ total number of parts

ratio L1 · (90, (2, 7), '$')

Share $90 in the ratio 2 : 7. Give both shares.

→ **$20 and $70** One part = amount ÷ total number of parts

ratio L1 · (150, (2, 3), '$')

Share $150 in the ratio 2 : 3. Give both shares.

→ **$60 and $90** One part = amount ÷ total number of parts

ratio L1 · (72, (5, 1), 'stickers')

Share 72 stickers in the ratio 5 : 1. Give both shares.

→ **60 and 12** One part = amount ÷ total number of parts

ratio L1 · (360, (7, 5), '$')

Share $360 in the ratio 7 : 5. Give both shares.

→ **$210 and $150** One part = amount ÷ total number of parts

ratio L1 · (48, (3, 5), 'marbles')

Share 48 marbles in the ratio 3 : 5. Give both shares.

→ **18 and 30** One part = amount ÷ total number of parts

ratio L1 · (500, (3, 2), '$')

Share $500 in the ratio 3 : 2. Give both shares.

→ **$300 and $200** One part = amount ÷ total number of parts

ratio L2 · (250, (3, 5), '$')

Share $250 in the ratio 3 : 5. Give both shares.

→ **$93.75 and $156.25** One part = amount ÷ total number of parts

ratio L2 · (360, (2, 3, 4), '$')

Share $360 in the ratio 2 : 3 : 4. Give all three shares.

→ **$80, $120 and $160** One part = amount ÷ total number of parts

ratio L2 · (95, (3, 2), '$')

Share $95 in the ratio 3 : 2. Give both shares.

→ **$57 and $38** One part = amount ÷ total number of parts

ratio L2 · (100, (3, 5), '$')

Share $100 in the ratio 3 : 5. Give both shares.

→ **$37.50 and $62.50** One part = amount ÷ total number of parts

ratio L2 · (630, (2, 3, 4), '$')

Share $630 in the ratio 2 : 3 : 4. Give all three shares.

→ **$140, $210 and $280** One part = amount ÷ total number of parts

ratio L2 · (2.4, (1, 2), 'kg')

Share 2.4 kg in the ratio 1 : 2. Give both shares.

→ **0.8 kg and 1.6 kg** One part = amount ÷ total number of parts

idx L1 · ('mult', 3, 'x', 4, 2, 3)

Simplify.

→ **6x^7** $a^m \times a^n = a^{m+n}$

$$3x^4 \times 2x^3$$

idx L1 · ('div', 12, 'a', 5, 4, 2)

Simplify.

→ **3a^3** $a^m \div a^n = a^{m-n}$

$$12a^5 \div 4a^2$$

idx L1 · ('pow', 2, 'k', 3, 2)

Simplify.

→ **4k^6** $(a^m)^n = a^{mn}$

$$(2k^3)^2$$

idx L1 · ('mult', 5, 'p', 2, 3, 6)

Simplify.

→ **15p^8** $a^m \times a^n = a^{m+n}$

$$5p^2 \times 3p^6$$

idx L1 · ('div', 20, 'n', 7, 5, 3)

Simplify.

→ **4n^4** $a^m \div a^n = a^{m-n}$

$$20n^7 \div 5n^3$$

idx L1 · ('mult', 4, 'a', 3, 1, 5)

Simplify.

→ **4a^8** $a^m \times a^n = a^{m+n}$

$$4a^3 \times a^5$$

idx L2 · ('mult', 4, 'x', 5, 5, 1)

Simplify.

→ **20x^6** $a^m \times a^n = a^{m+n}$

$$4x^5 \times 5x$$

idx L2 · ('div', 15, 'm', 8, 5, 3)

Simplify.

→ **3m^5** $a^m \div a^n = a^{m-n}$

$$15m^8 \div 5m^3$$

idx L2 · ('pow', 3, 'a', 2, 3)

Simplify.

→ **27a^6** $(a^m)^n = a^{mn}$

$$(3a^2)^3$$

idx L2 · ('div', 18, 'y', 6, 6, 1)

Simplify.

→ **3y^5** $a^m \div a^n = a^{m-n}$

$$18y^6 \div 6y$$

idx L2 · ('mult', 2, 'a', 3, 7, 4)

Simplify.

→ **14a^7** $a^m \times a^n = a^{m+n}$

$$2a^3 \times 7a^4$$

idx L2 · ('pow', 5, 'x', 4, 2)

Simplify.

→ **25x^8** $(a^m)^n = a^{mn}$

$$(5x^4)^2$$

idx L2 · ('div', 24, 'p', 9, 8, 4)

Simplify.

→ **3p^5** $a^m \div a^n = a^{m-n}$

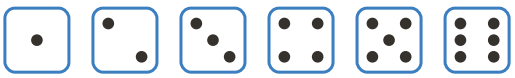
$$24p^9 \div 8p^4$$

E STRAIGHT-LINE GRAPHS, DATA AND PROBABILITY

freq L1 · ('die', 300, 'a 4', 1)

A fair die is rolled 300 times. How many times would you expect a 4?

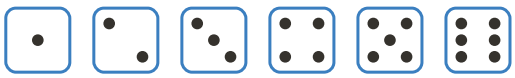
→ **50** Expected frequency = P(E) × number of trials



freq L1 · ('die', 120, 'an even number', 3)

A fair die is rolled 120 times. How many times would you expect an even number?

→ **60** Expected frequency = P(E) × number of trials



freq L1 · ('coin', 50)

A fair coin is tossed 50 times. How many heads would you expect?

→ **25** Expected frequency = P(E) × number of trials

freq L1 · ('p', 'late', '0.2', 40)

On how many days would you expect the bus to be late?

→ **8** Expected frequency = P(E) × number of trials

P(the bus is late)	0.2
School days	40

freq L1 · ('rel', 42, 100, 'heads')

Heads came up 42 times in 100 tosses. Find the relative frequency of heads.

→ **0.42 (or 21/50)** Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$

freq L1 · ('tab', [('Red', 12), ('Blue', 18), ('Green', 20)], 'blue')

A spinner was spun 50 times. Find the relative frequency of blue.

→ **0.36 (or 9/25)** Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$

Colour	Red	Blue	Green
Frequency	12	18	20

freq L1 · ('rel', 15, 60, 'sixes')

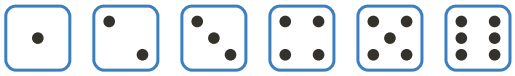
A six came up 15 times in 60 rolls. Find the relative frequency of a six.

→ **0.25 (or 1/4)** Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$

freq L1 · ('die', 600, 'a number greater than 4', 2)

A fair die is rolled 600 times. How many times would you expect a number greater than 4?

→ **200** Expected frequency = $P(\text{E}) \times$ number of trials



freq L1 · ('rel', 27, 90, 'heads')

Heads came up 27 times in 90 tosses. Find the relative frequency of heads.

→ **0.3 (or 3/10)** Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$

freq L2 · ('p', 'rain', '0.15', 60)

On how many days would you expect it to rain?

→ **9** Expected frequency = $P(\text{E}) \times$ number of trials

P(it rains)	0.15
Days	60

freq L2 · ('p', 'late', '0.35', 400)

On how many days would you expect the bus to be late?

→ **140** Expected frequency = $P(\text{E}) \times$ number of trials

P(the bus is late)	0.35
School days	400

freq L2 · ('rel', 7, 40, 'heads')

Heads came up 7 times in 40 tosses. Find the relative frequency of heads.

→ **0.175 (or 7/40)** Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$

freq L2 · ('tab', [('Red', 14), ('Blue', 26), ('Green', 10)], 'red')

A spinner was spun 50 times. Find the relative frequency of red.

→ **0.28 (or 7/25)** Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$

Colour	Red	Blue	Green
Frequency	14	26	10

freq L2 · ('rel', 33, 150, 'sixes')

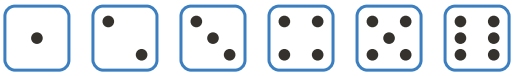
A six came up 33 times in 150 rolls. Find the relative frequency of a six.

→ **0.22 (or 11/50)** Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$

freq L2 · ('die', 450, 'a 1 or a 2', 2)

A fair die is rolled 450 times. How many times would you expect a 1 or a 2?

→ **150** Expected frequency = $P(E) \times$ number of trials



freq L2 · ('tab', [('Red', 23), ('Blue', 41), ('Green', 36)], 'green')

A spinner was spun 100 times. Find the relative frequency of green.

→ **0.36 (or 9/25)** Relative frequency = $\frac{\text{frequency}}{\text{number of trials}}$

Colour	Red	Blue	Green
Frequency	23	41	36

linear L1 · ('gi', 3, -2, 'mc')

State the gradient and the y-intercept.

→ **gradient 3, y-intercept −2** $y = mx + c$

$$y = 3x - 2$$

linear L1 · ('gi', 2, 5, 'mc')

State the gradient and the y-intercept.

→ **gradient 2, y-intercept 5** $y = mx + c$

$$y = 2x + 5$$

linear L1 · ('gi', -4, 1, 'mc')

State the gradient and the y-intercept.

→ **gradient −4, y-intercept 1** $y = mx + c$

$$y = -4x + 1$$

linear L2 · ('gi', -2, 7, 'mc')

State the gradient and the y-intercept.

→ **gradient −2, y-intercept 7** $y = mx + c$

$$y = -2x + 7$$

linear L2 · ('gi', 0.5, -3, 'mc')

State the gradient and the *y*-intercept.

→ **gradient 0.5, y-intercept −3** $y = mx + c$

$$y = 0.5x - 3$$

linear L2 · ('gi', -2, 6, 'cm')

State the gradient and the *y*-intercept.

→ **gradient −2, y-intercept 6** $y = mx + c$

$$y = 6 - 2x$$

mean L1 · ('ft', (1, 2, 3, 4, 5), (1, 1, 2, 4, 7))

Find the mean of these scores.

→ **4** Mean = $\frac{\Sigma fx}{\Sigma f}$

Score	1	2	3	4	5
Frequency	1	1	2	4	7

mean L1 · ('ft', (0, 1, 2, 3, 4), (2, 5, 3, 1, 4))

Find the mean of these scores.

→ **2** Mean = $\frac{\Sigma fx}{\Sigma f}$

Score	0	1	2	3	4
Frequency	2	5	3	1	4

mean L1 · ('ft', (2, 3, 4, 5, 6), (4, 7, 2, 1, 7))

Find the mean of these scores.

→ **4** Mean = $\frac{\Sigma fx}{\Sigma f}$

Score	2	3	4	5	6
Frequency	4	7	2	1	7

mean L1 · ('ft', (5, 6, 7, 8, 9), (4, 5, 5, 1, 6))

Find the mean of these scores.

→ **7** Mean = $\frac{\Sigma fx}{\Sigma f}$

Score	5	6	7	8	9
Frequency	4	5	5	1	6

mean L2 · ('ft', (1, 2, 3, 4, 5), (3, 6, 8, 5, 2))

Find the mean of these scores, correct to 1 decimal place.

→ **2.9** Mean = $\frac{\Sigma fx}{\Sigma f}$

Score	1	2	3	4	5
Frequency	3	6	8	5	2

mean L2 · ('ft', (2, 3, 4, 5, 6), (2, 5, 9, 6, 3))

Find the mean of these scores, correct to 1 decimal place.

→ **4.1** Mean = $\frac{\Sigma fx}{\Sigma f}$

Score	2	3	4	5	6
Frequency	2	5	9	6	3

mean L2 · ('ft', (5, 6, 7, 8, 9), (3, 4, 6, 5, 4))

Find the mean of these scores, correct to 1 decimal place.

→ **7.1** Mean = $\frac{\Sigma fx}{\Sigma f}$

Score	5	6	7	8	9
Frequency	3	4	6	5	4

mean L2 · ('ft', (10, 11, 12, 13, 14), (2, 6, 4, 5, 1))

Find the mean of these scores, correct to 1 decimal place.

→ **11.8** Mean = $\frac{\Sigma fx}{\Sigma f}$

Score	10	11	12	13	14
Frequency	2	6	4	5	1